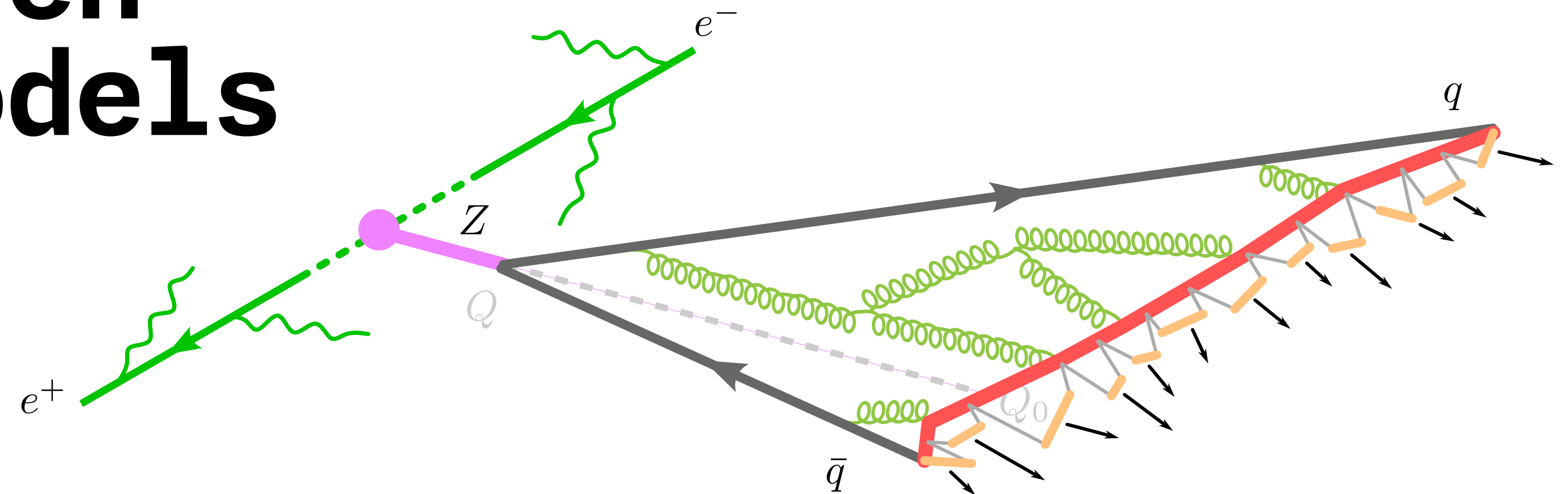


Tuning data-driven hadronization models

a.k.a “Machine learning tuning”

LPCC MCWG Tuning Forum

September 22nd, 2026



Based on work with MLHAD

Tony Menzo

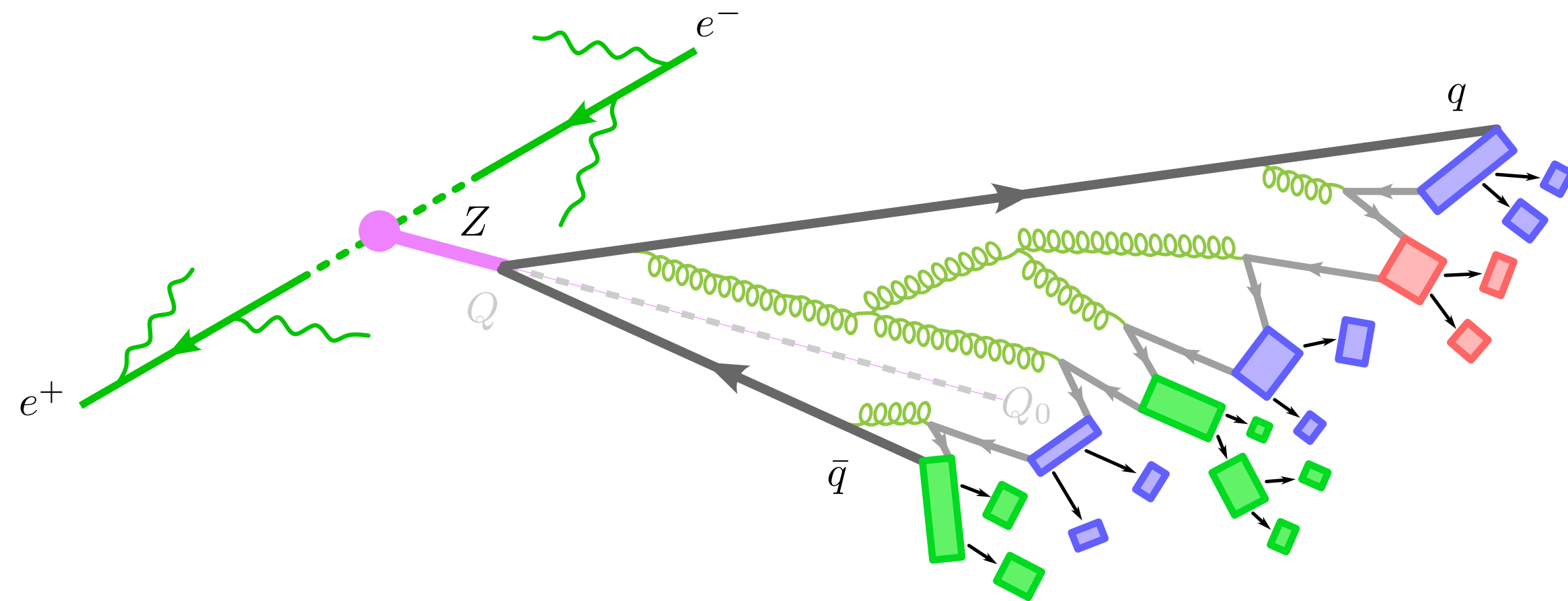
Joint postdoc @  Fermilab and the University of  Alabama

Outline

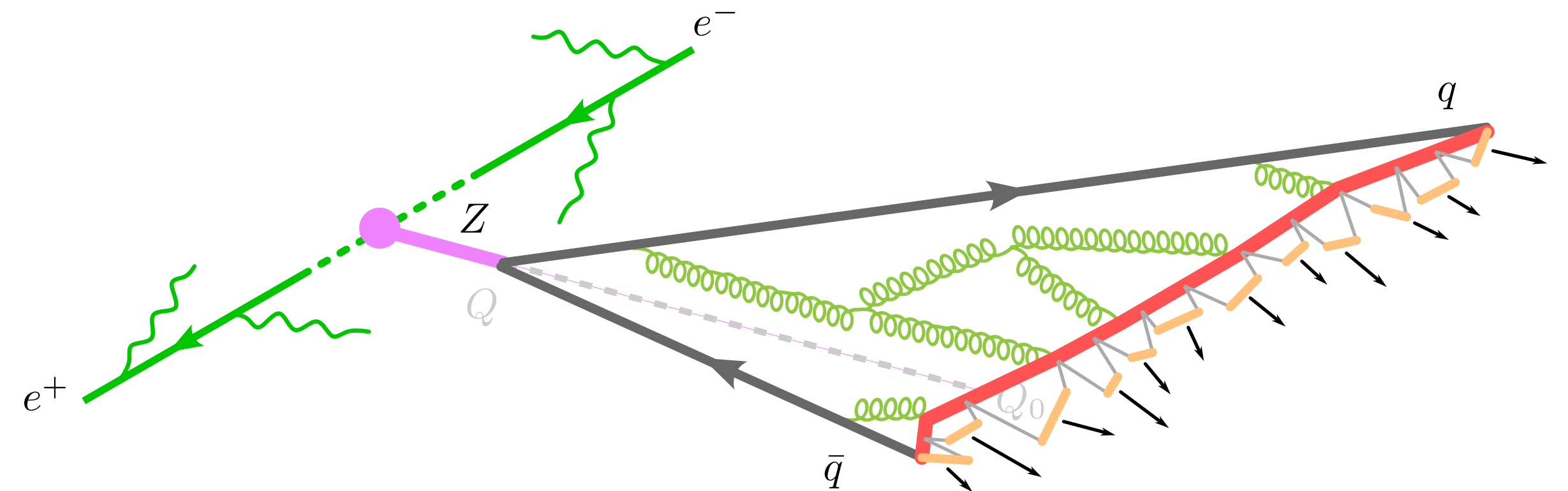
1. (String) Hadronization as an inverse problem
 - The Lund “saddle point”
2. Data-driven hadronization models
 - **HOMER, iHOMER, MAGIC, ...**
3. Conclusions

Hadronization (speed-run)

Converting colored **partons** to color-singlet hadrons



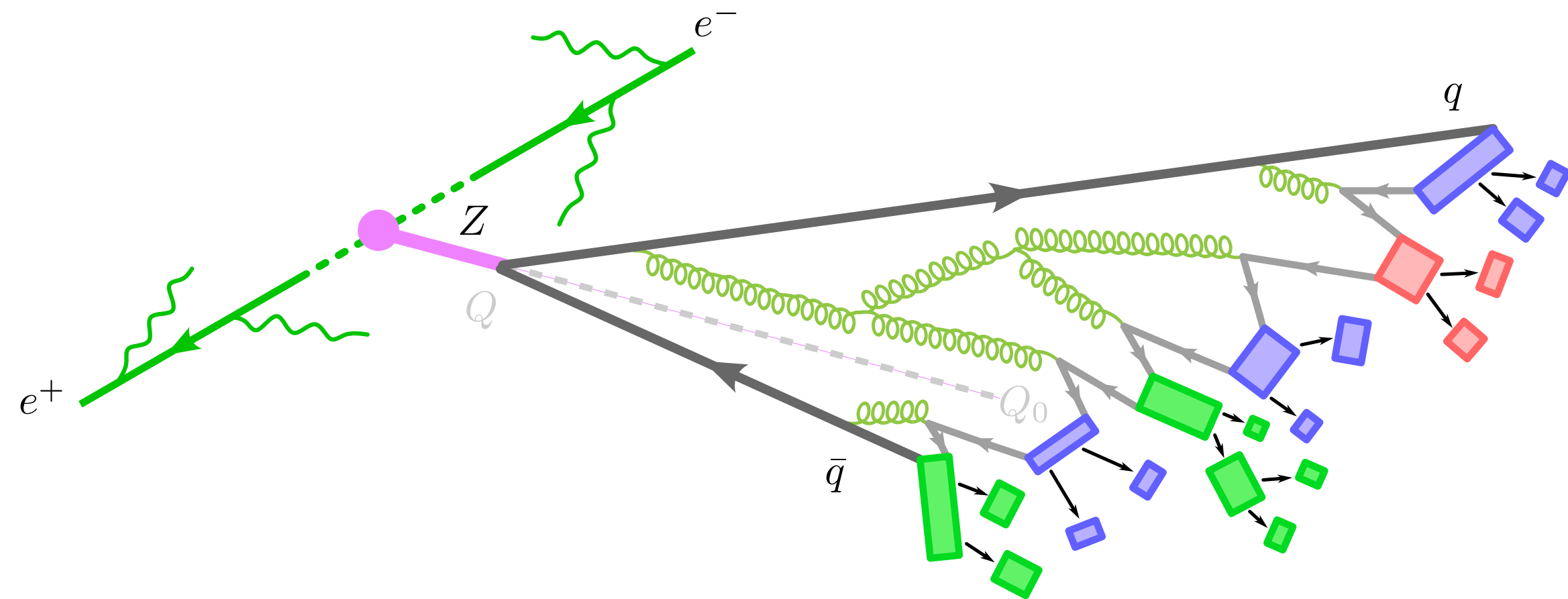
Cluster



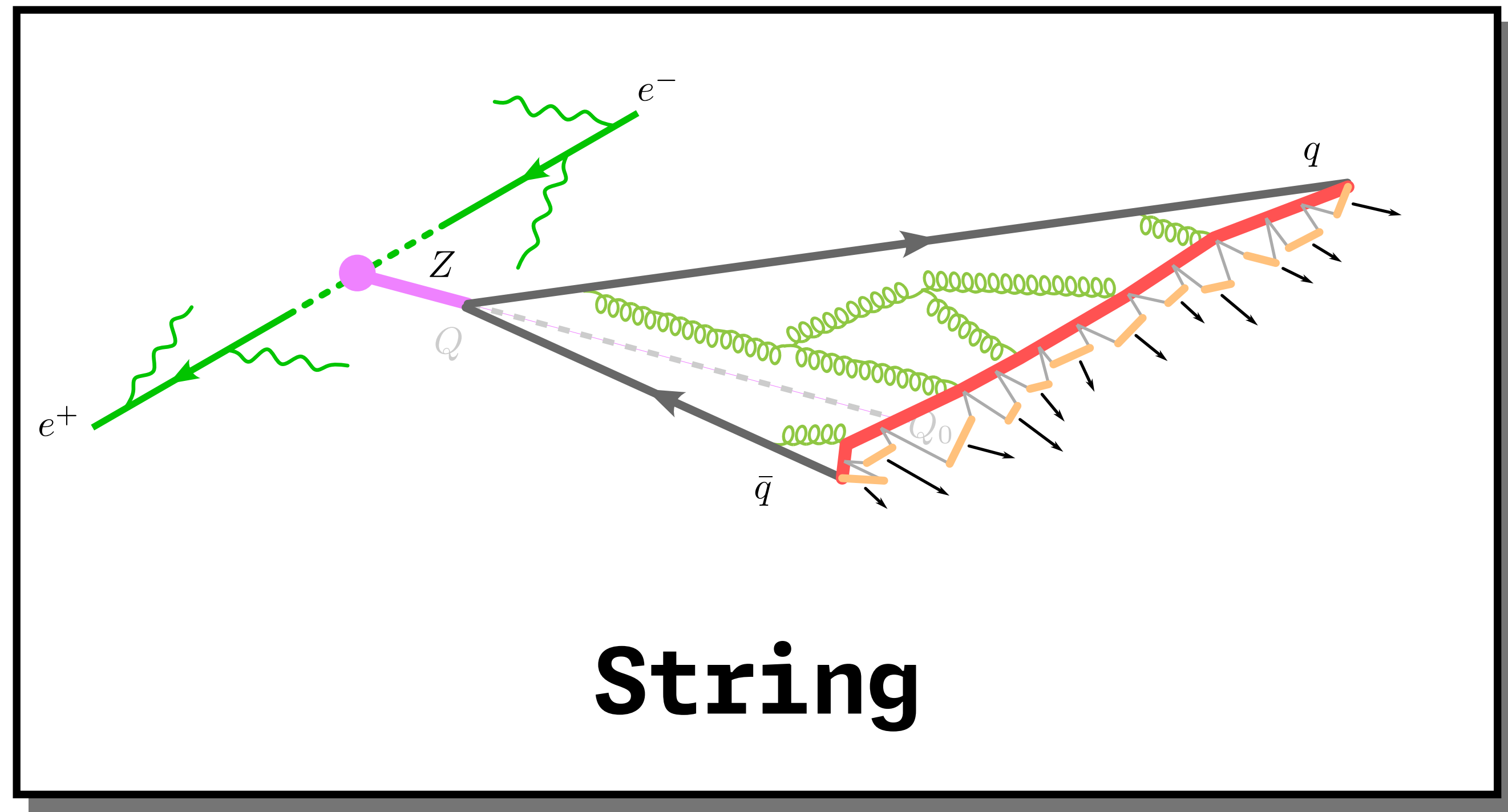
String

Hadronization (speed-run)

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Cluster



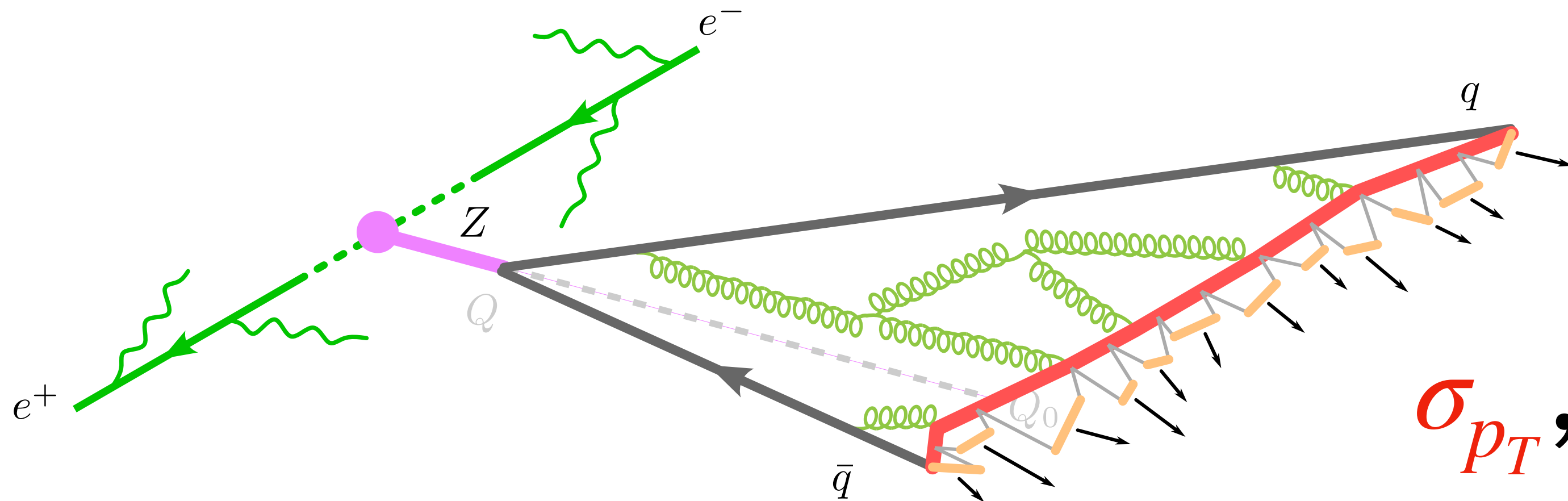
String

String Hadronization (speed-run)

Converting colored **partons** to color-singlet **hadrons**

Lund string model

~“core” params:



a, b } Longitudinal consistency
causality +
left/right symmetry

$\sigma_{p_T}, \rho \equiv P_{u,d}/P_s$ } Schwinger pair
production

ξ, x, y } Baryon (diquark)
production

r_b, r_c } Heavy endpoints

A General Model for Jet Fragmentation

B. Andersson, G. Gustafson, B. Söderberg¹

Department of Theoretical Physics, University of Lund, Sölvegatan 14A, S-22362 Lund, Sweden

Received 14 April 1983

String Hadronization (speed-run)

Converting colored **partons** to color-singlet **hadrons**

1. Flavor selection ρ, ξ, x, y
+ **hadron spectroscopy**

2. String break $\vec{k}_\perp \sim \mathcal{N}_2(0, \sigma_{p_T} I)$

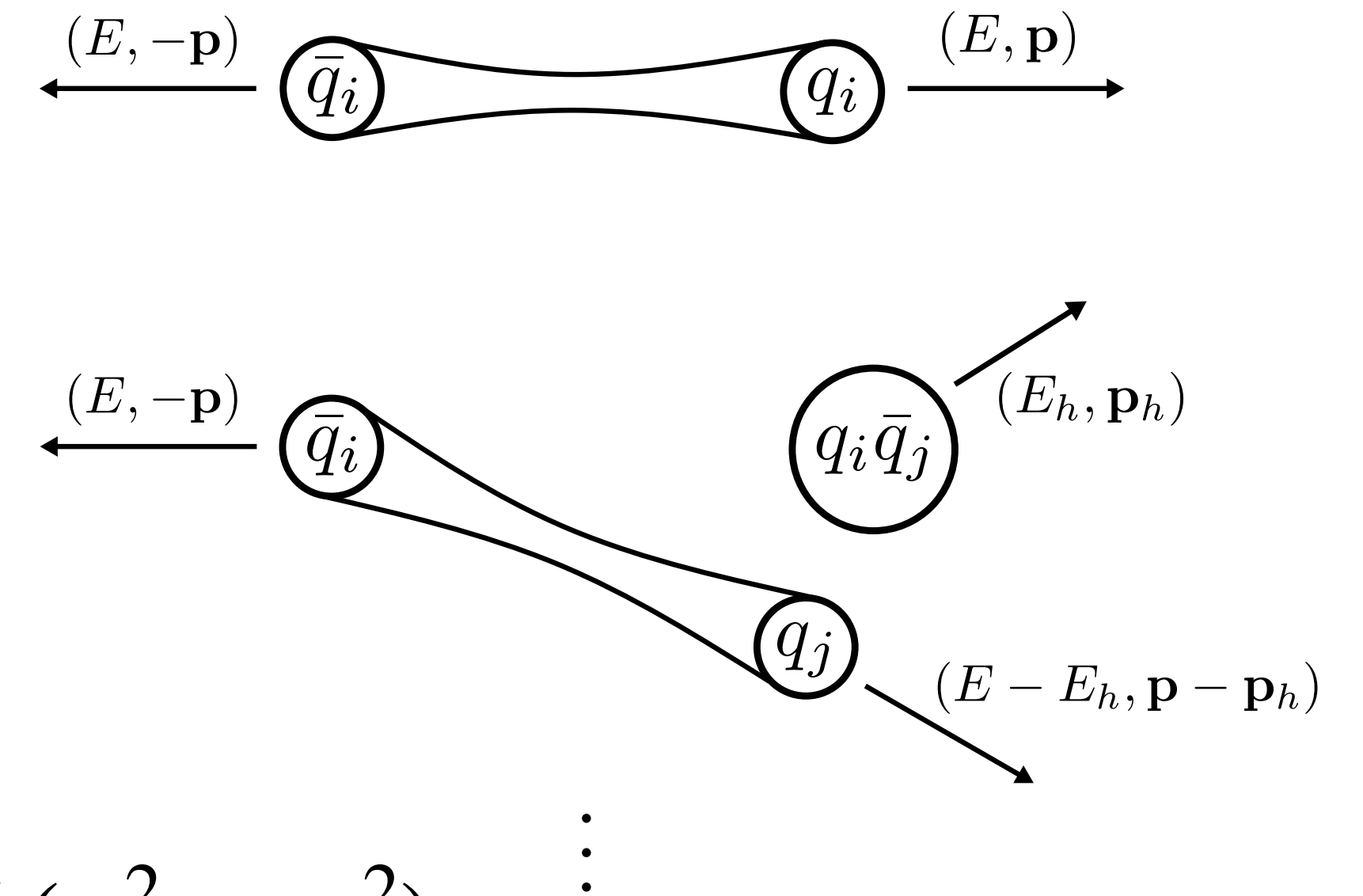
3. Hadron kinematics

$$f(z | m_T^2, M^2) \propto \frac{(1-z)^A}{z^C} \exp \left[-\frac{B}{z} \right]$$

$$A \equiv a_j, \quad B \equiv b(p_T^2 + m_h^2),$$

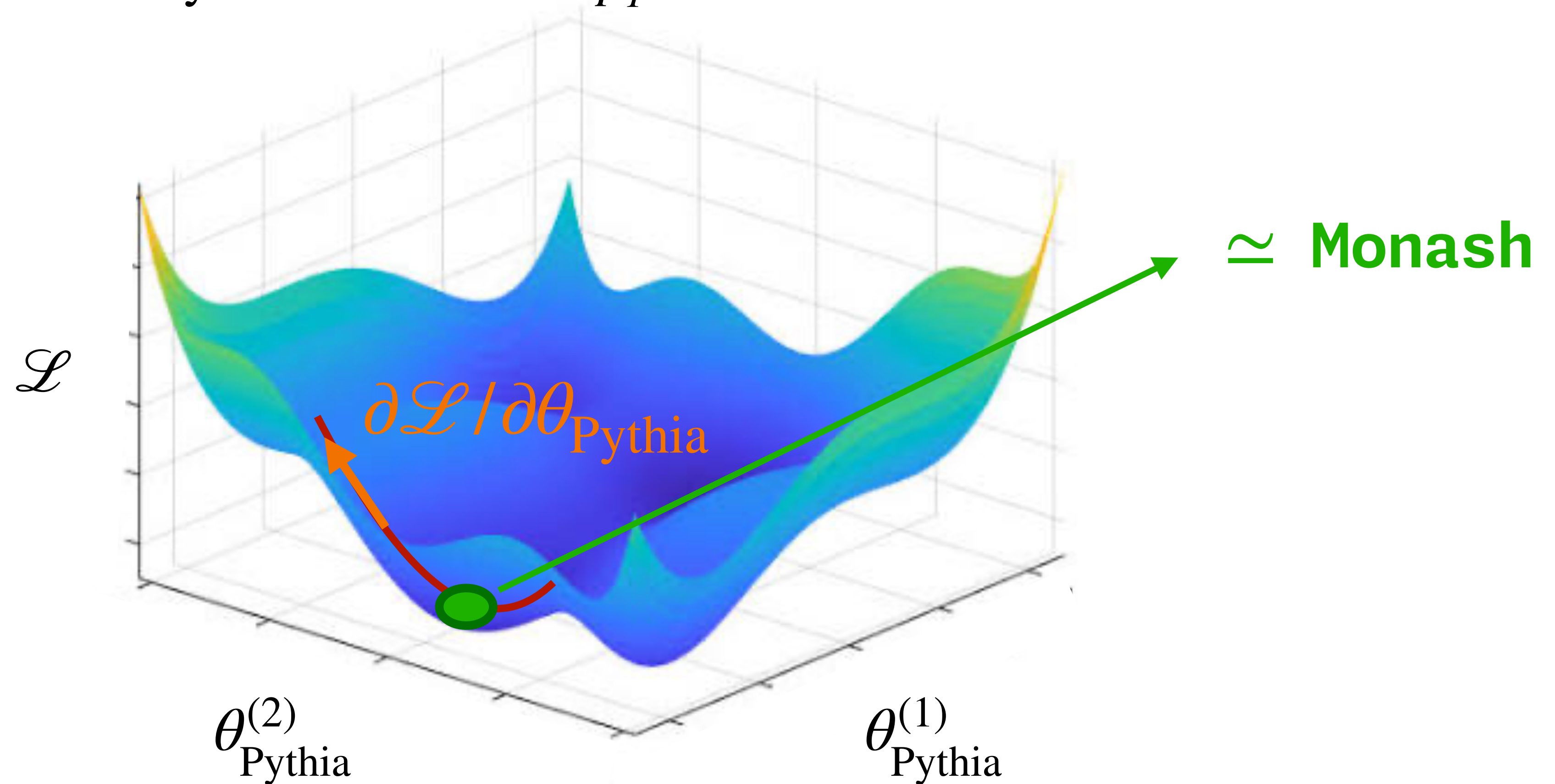
$$C \equiv 1 + (a_j - a_i) + r_j b m_h^2$$

$q\bar{q}$ -string



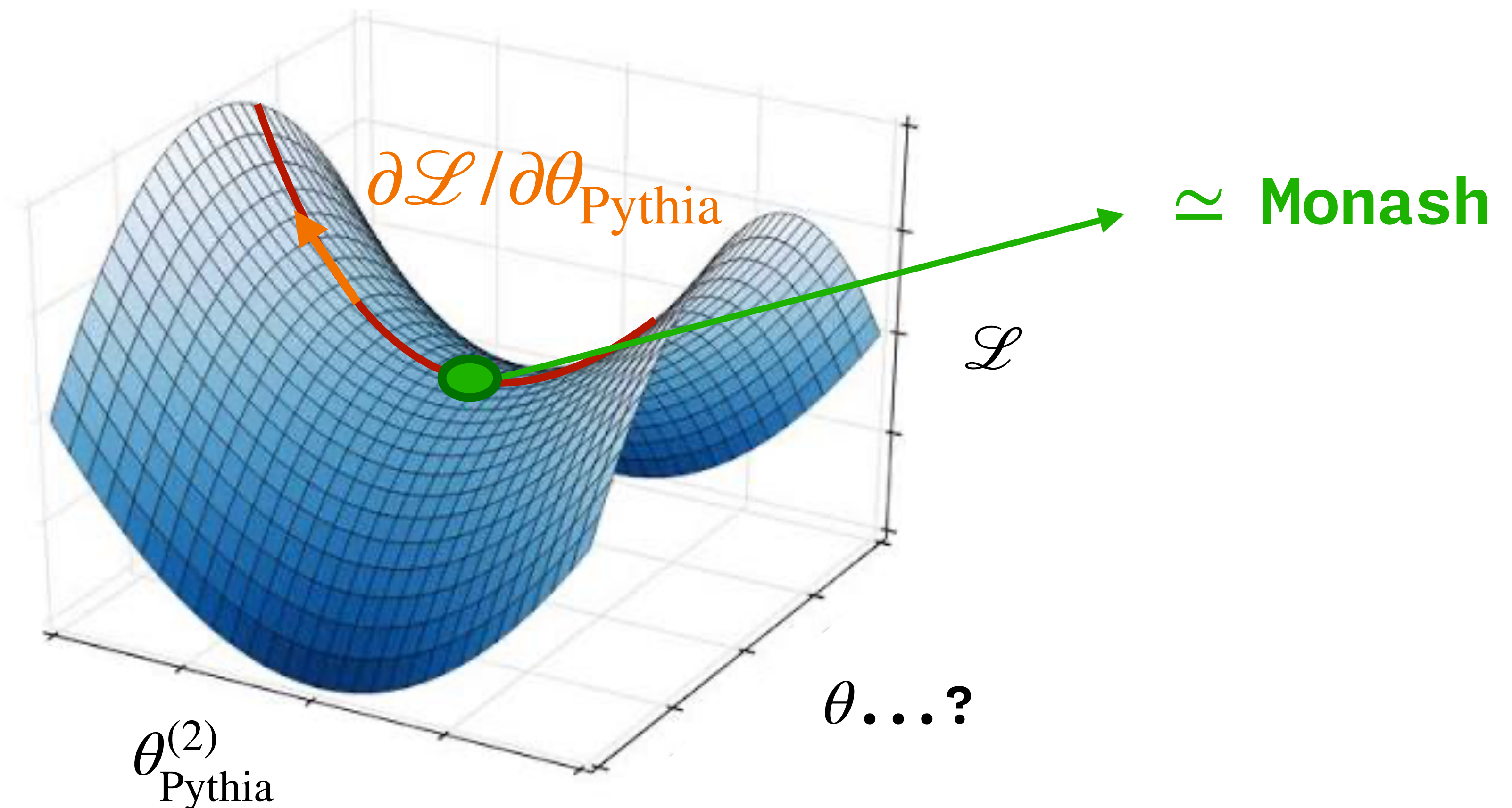
String Hadronization (speed-run)

$$\mathcal{L} \simeq | \mathcal{O}_{\text{sim}}(\theta_{\text{Pythia}} \equiv \{a, b, \sigma_{p_T}, \rho, \xi, x, y, \dots\}) - \mathcal{O}_{\text{data}} |$$



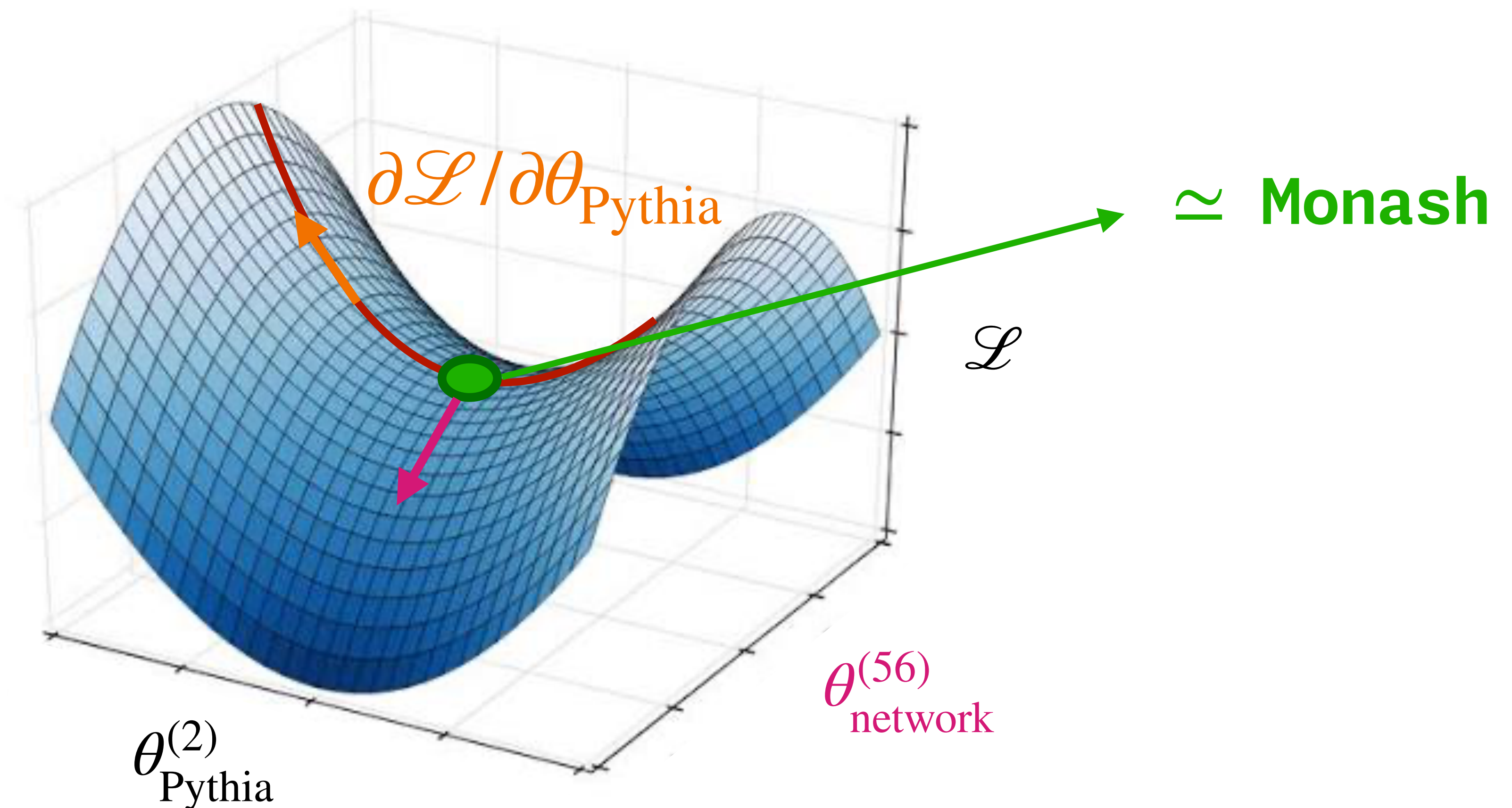
String Hadronization (speed-run)

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Data-driven hadronization models

$$\mathcal{O}_{\text{Pythia}}(\theta_{\text{Pythia}} \equiv \{a, b, \sigma_{p_T}, \rho, \xi, x, y, \dots\}) \rightarrow \mathcal{O}_{\text{ML surrogate}}(\vec{\theta}_{\text{network}})$$



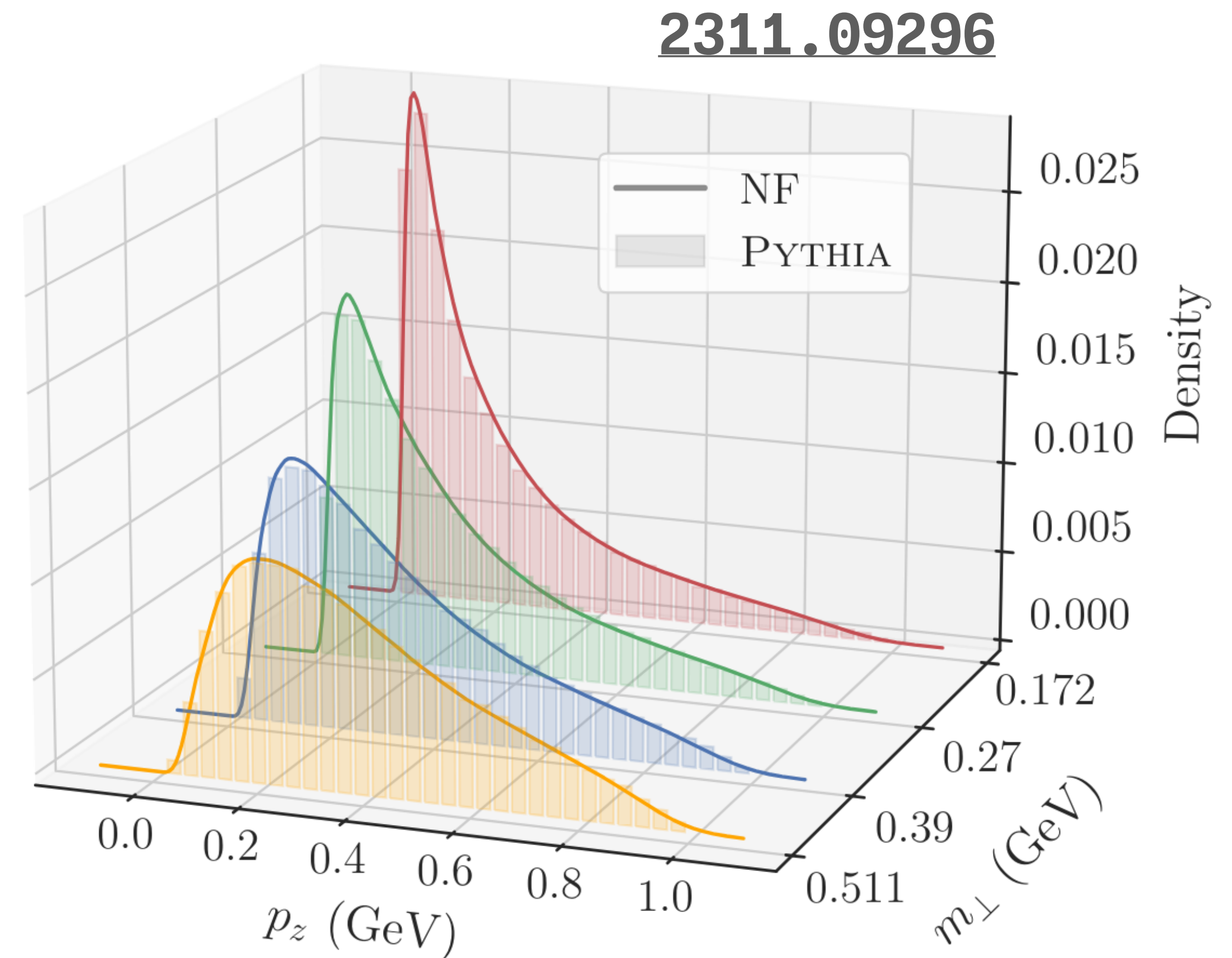
Data-driven hadronization models

MAGIC

A couple of nice ways to do this...

Start from saddle point, i.e. either 1. train network to replicate micro string model physics

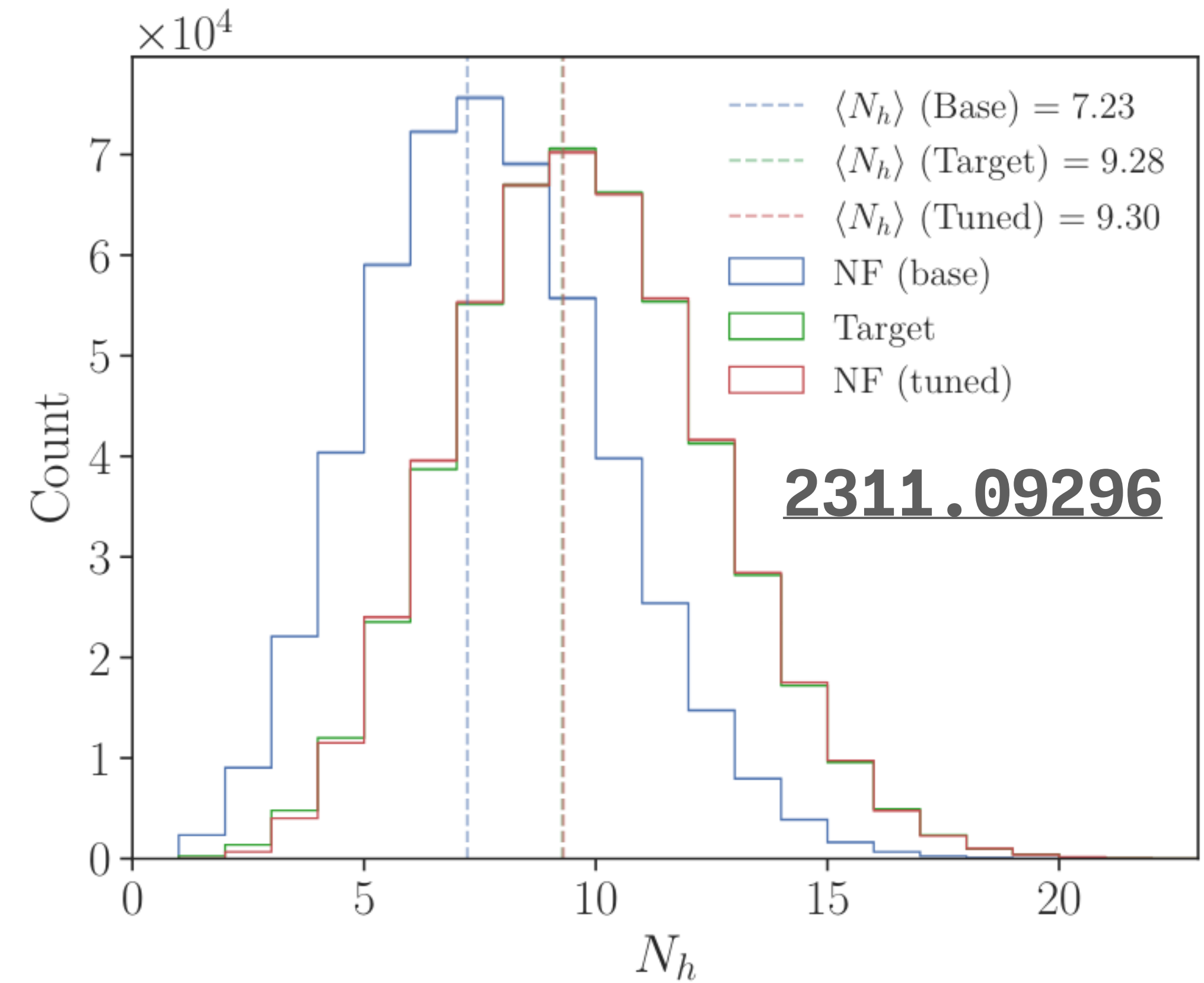
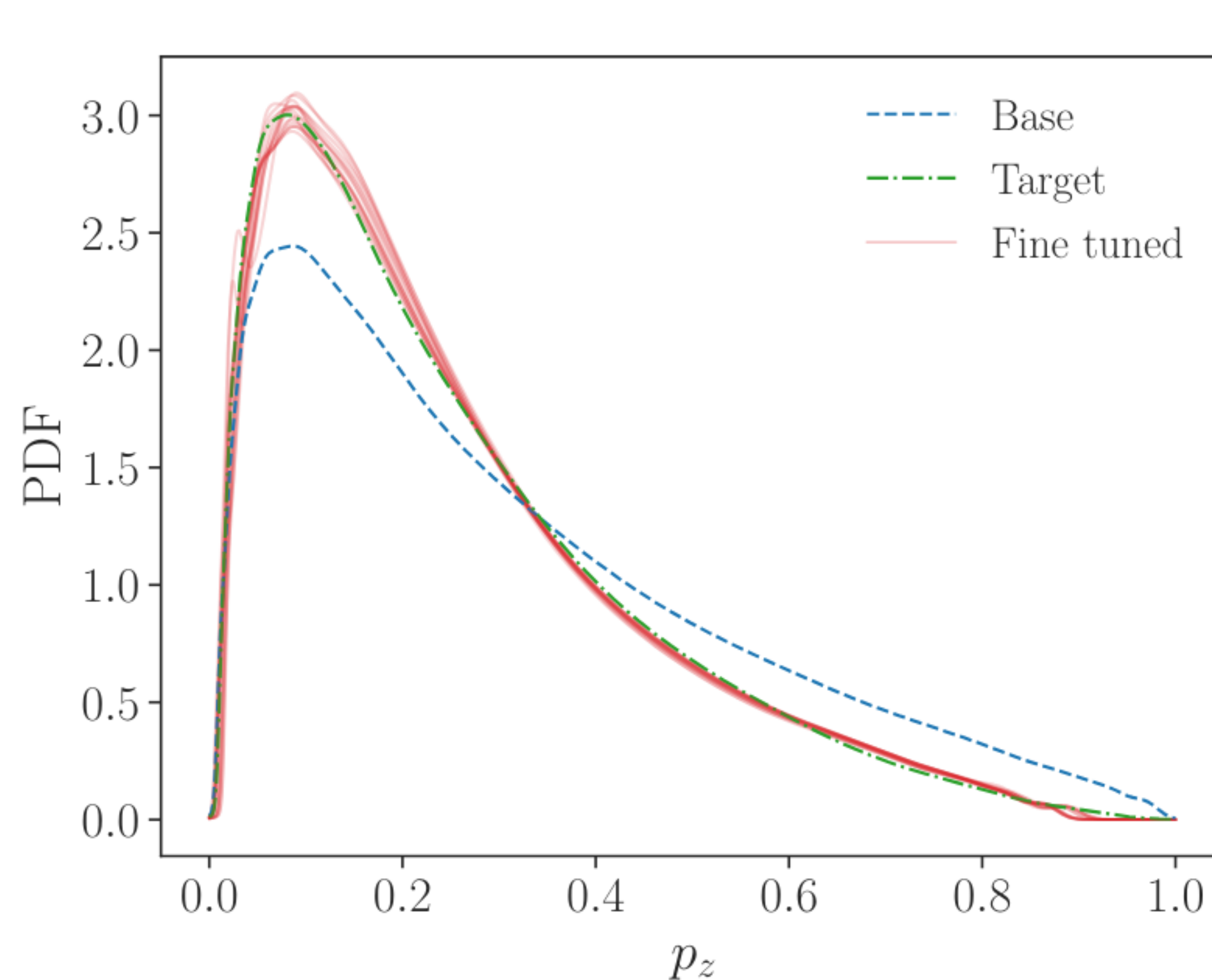
$$\mathcal{O}_{\text{Pythia}}(\theta_{\text{Pythia}}) \simeq \mathcal{O}_{\text{ML surrogate}}(\vec{\theta}_{\text{network}})$$



Data-driven hadronization models

MAGIC

... and let θ_{network} float to minimize $|\mathcal{O}_{\text{network}} - \mathcal{O}_{\text{data}}|$



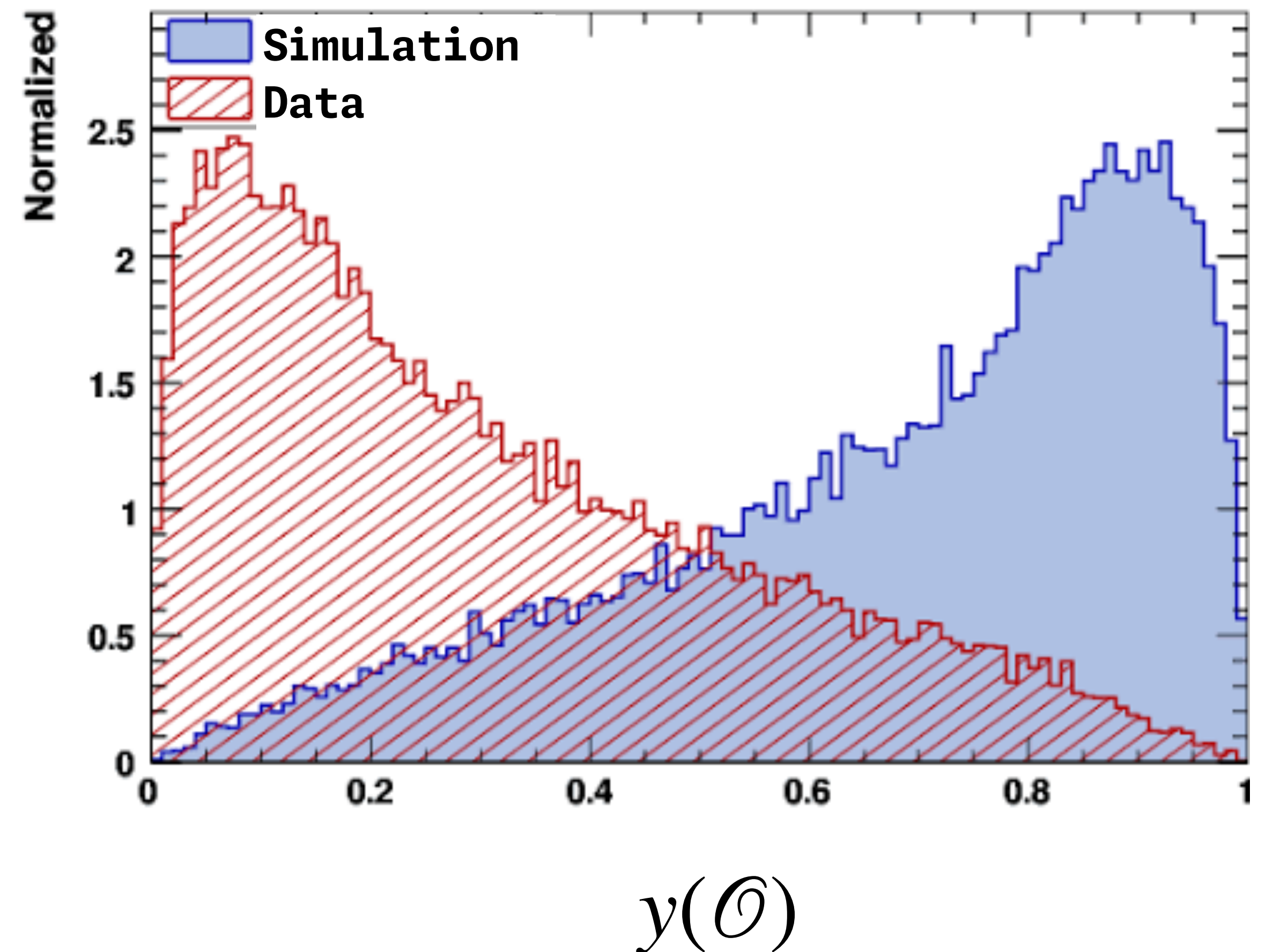
Data-driven hadronization models

HOMER ([2410.06342](#))

A couple of nice ways to do this...

Start from saddle point,
i.e. either 2. train
network to distinguish
between sim and data

$$y(\mathcal{O}_{\text{Pythia}}(\theta_{\text{Pythia}})) \rightarrow 0, \quad y(\mathcal{O}_{\text{data}}) \rightarrow 1$$

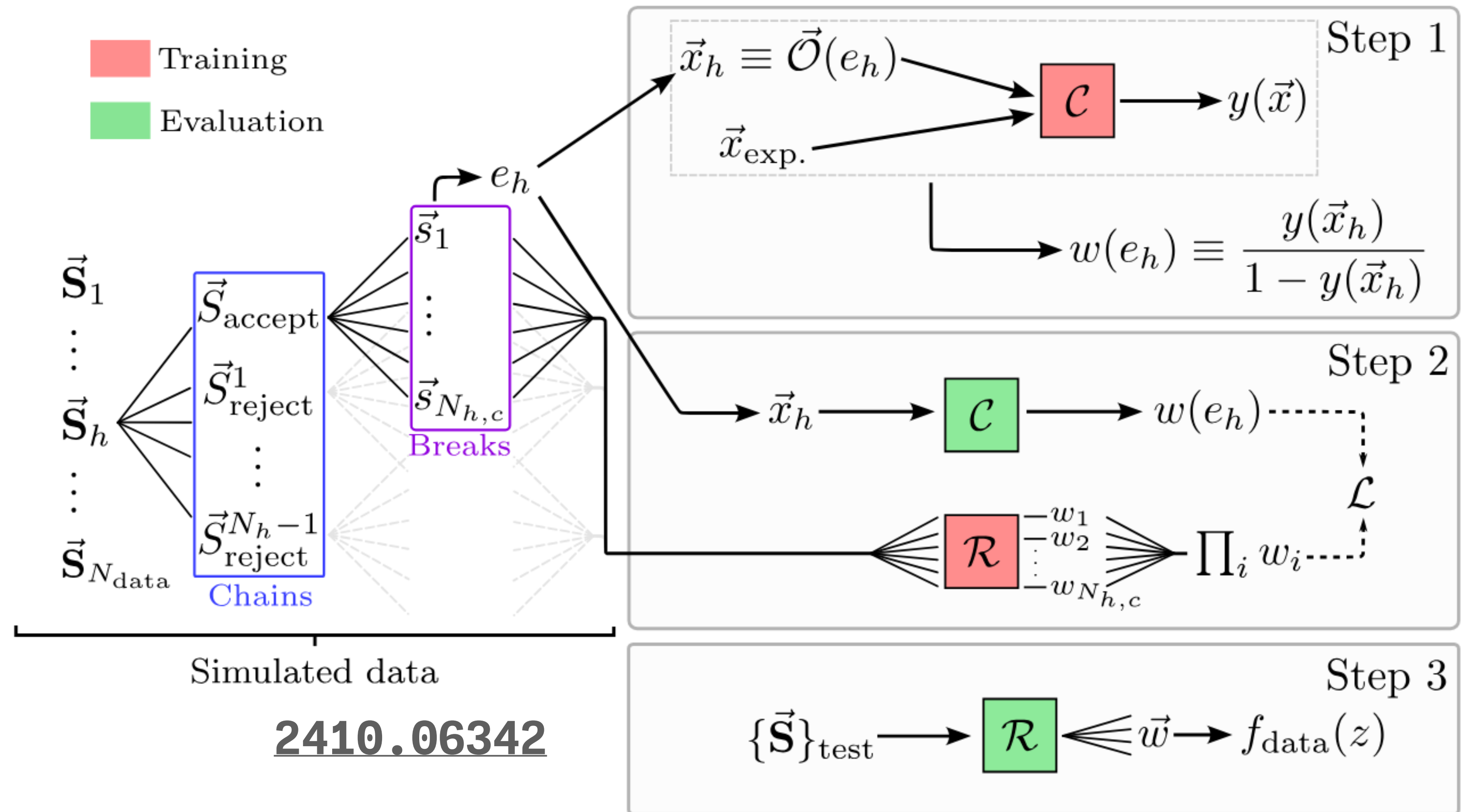


Data-driven hadronization models

HOMER ($q\bar{q}$)

Distribute classifier weight amongst string breaks to minimize

$$w_{\text{classifier}} - \prod_i w_{\text{network}}^{(i)}$$

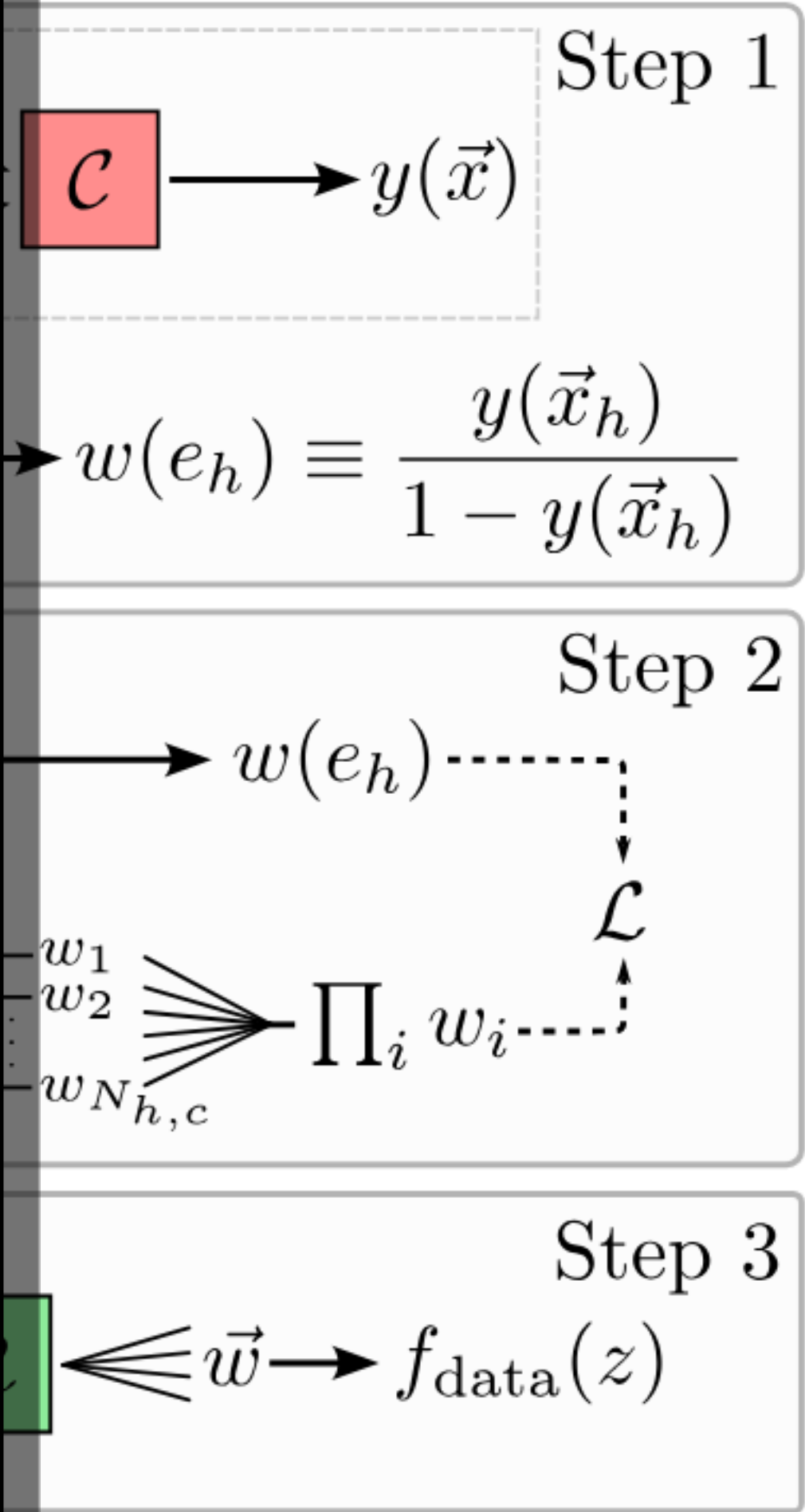
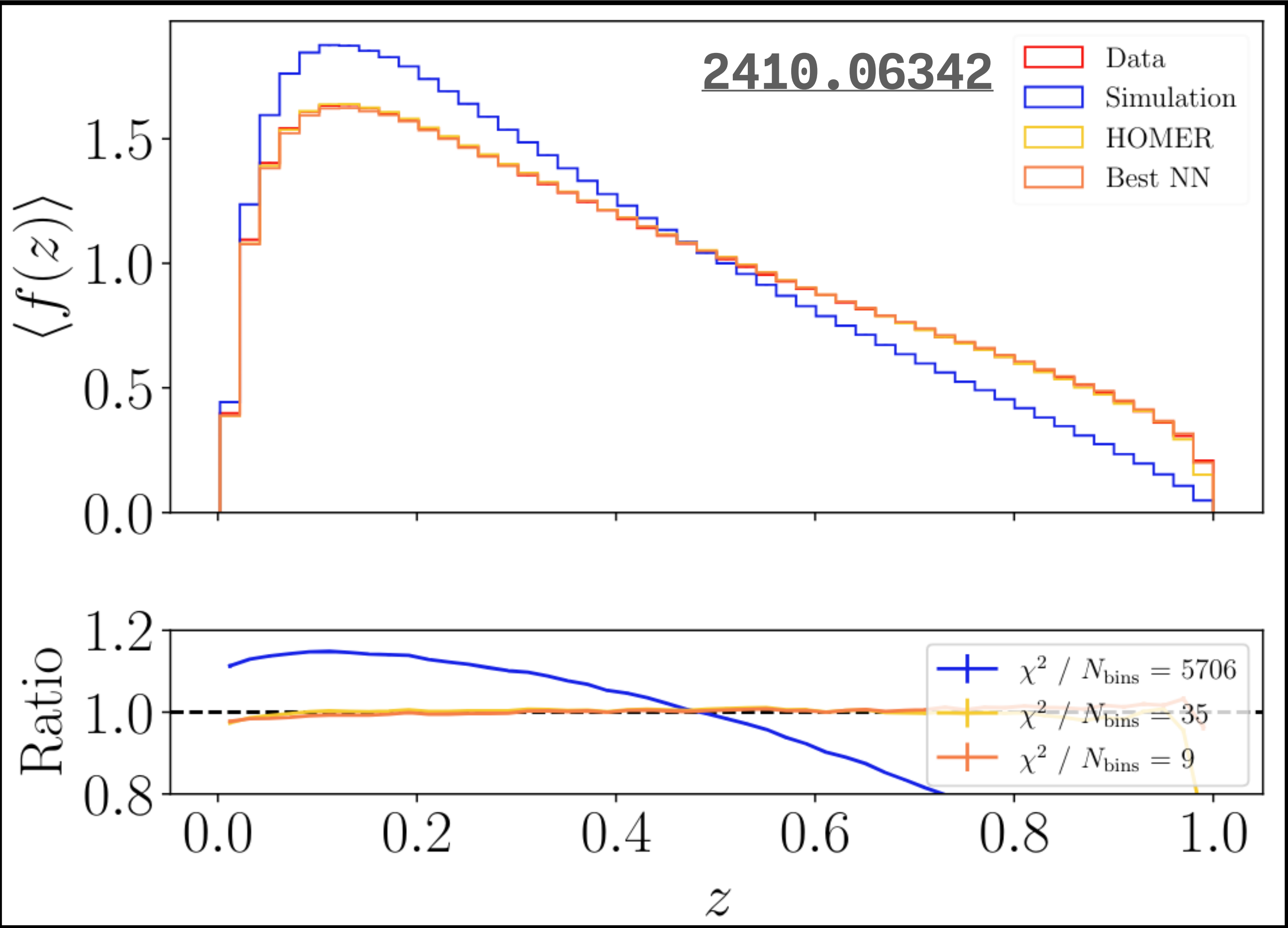


Data-driven hadronization models

HOMER ($q\bar{q}$)

Distribute classifier weight amongs string breaks minimize

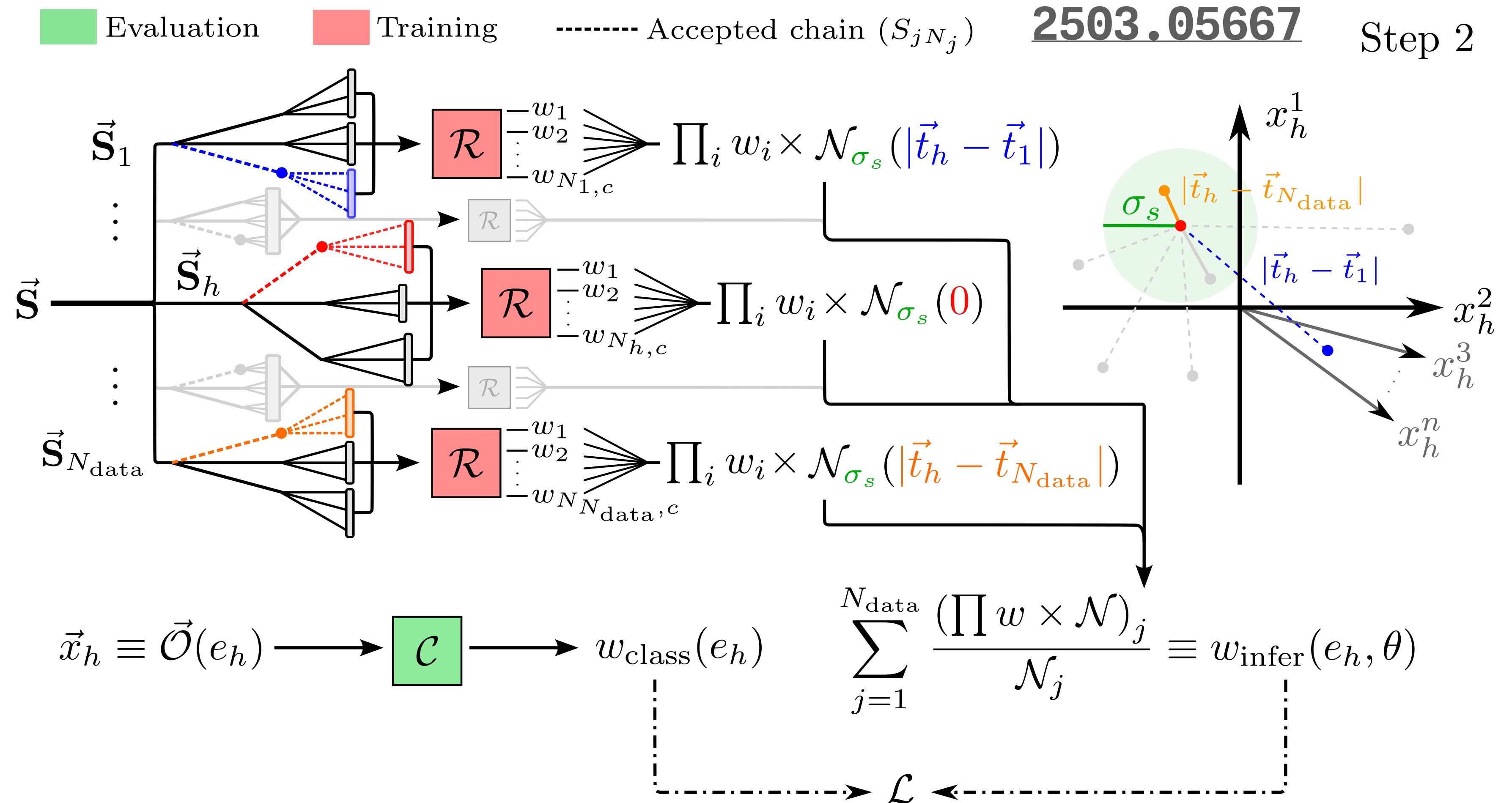
$$w_{\text{classifier}} = \prod_i w_{\text{ne}}$$



Data-driven hadronization models

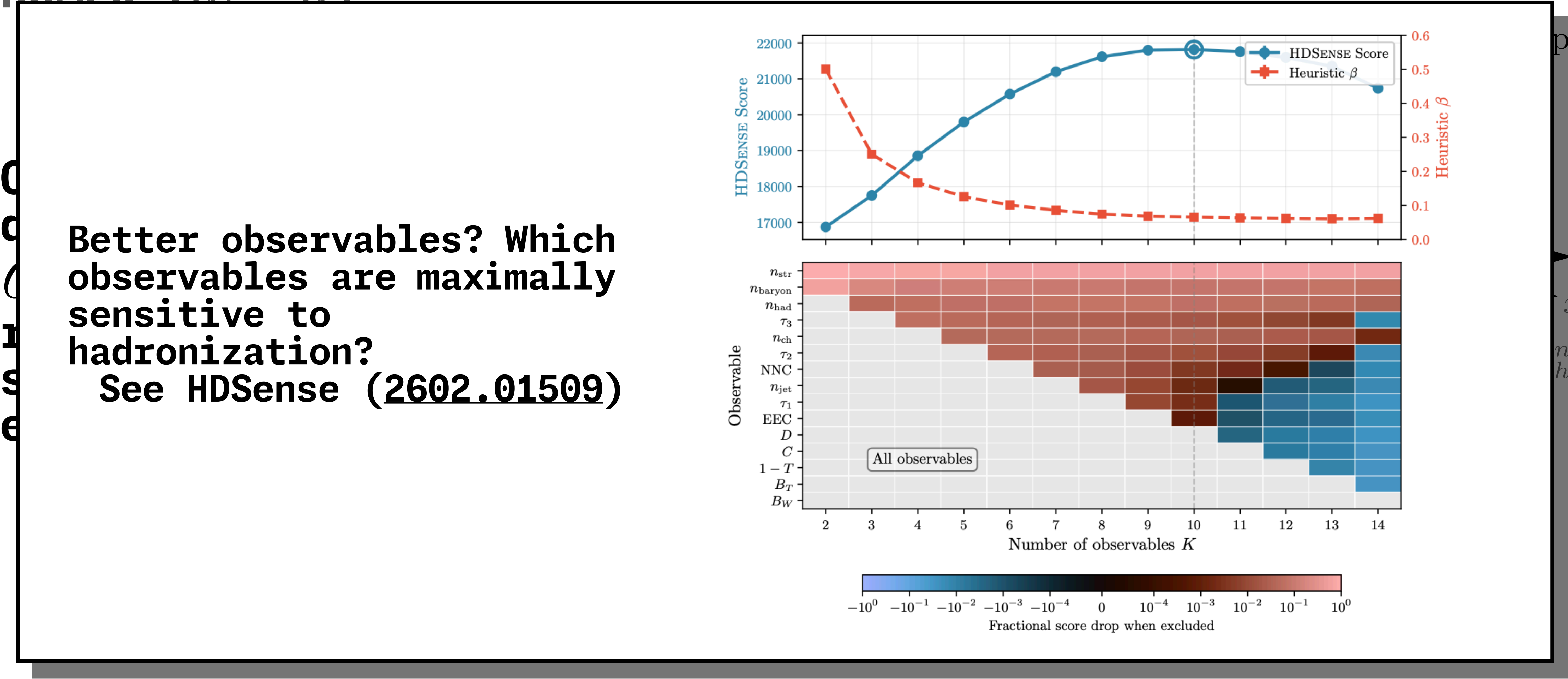
HOMER ($qg\cdots\bar{q}$)

Observable degeneracies
 $\mathcal{O}(S_1) \simeq \mathcal{O}(S_2)$
requires smearing in event space



Data-driven hadronization models

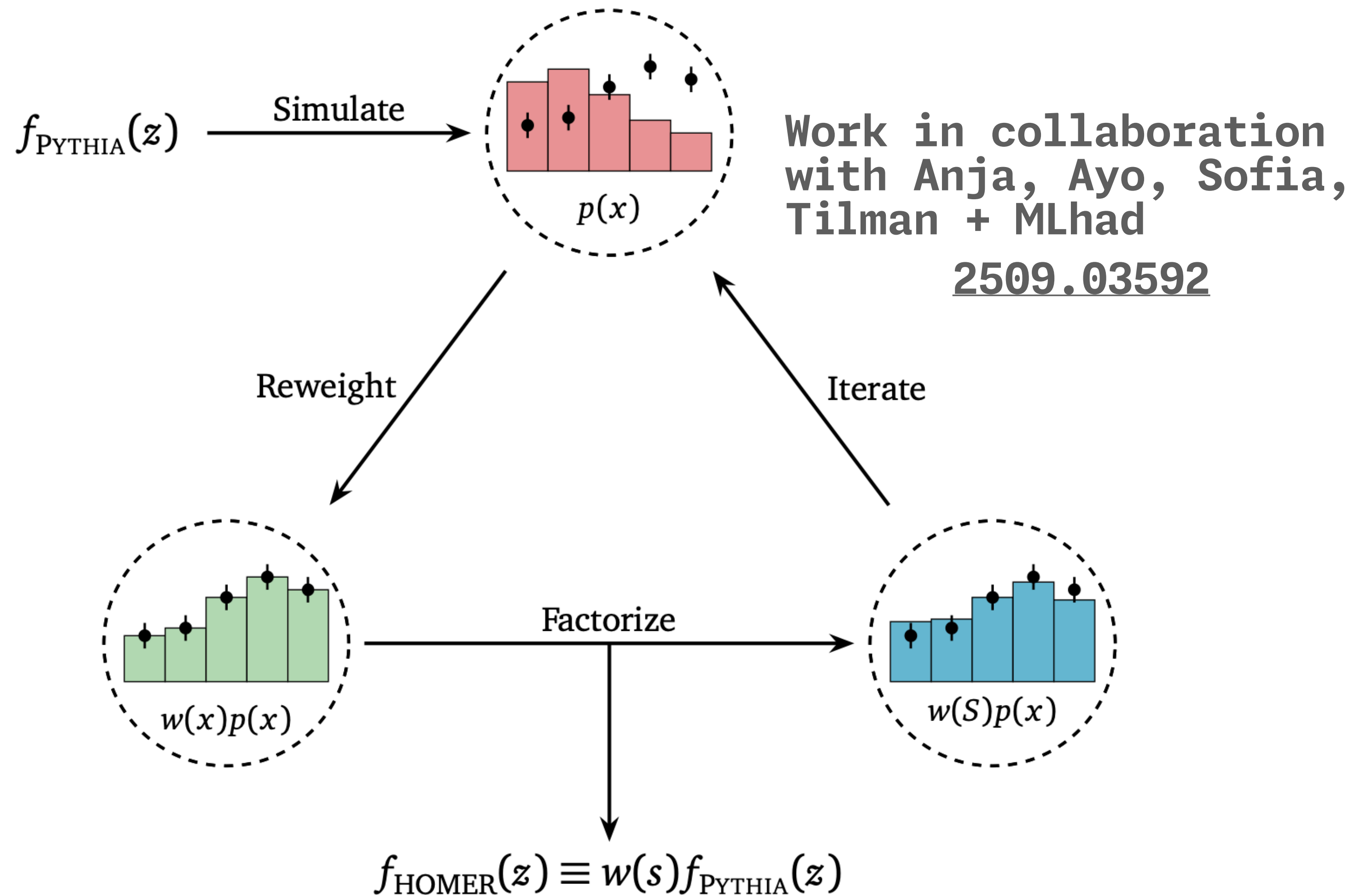
HOMER ($q\bar{q}\dots\bar{q}$)



Data-driven hadronization models

iHOMER

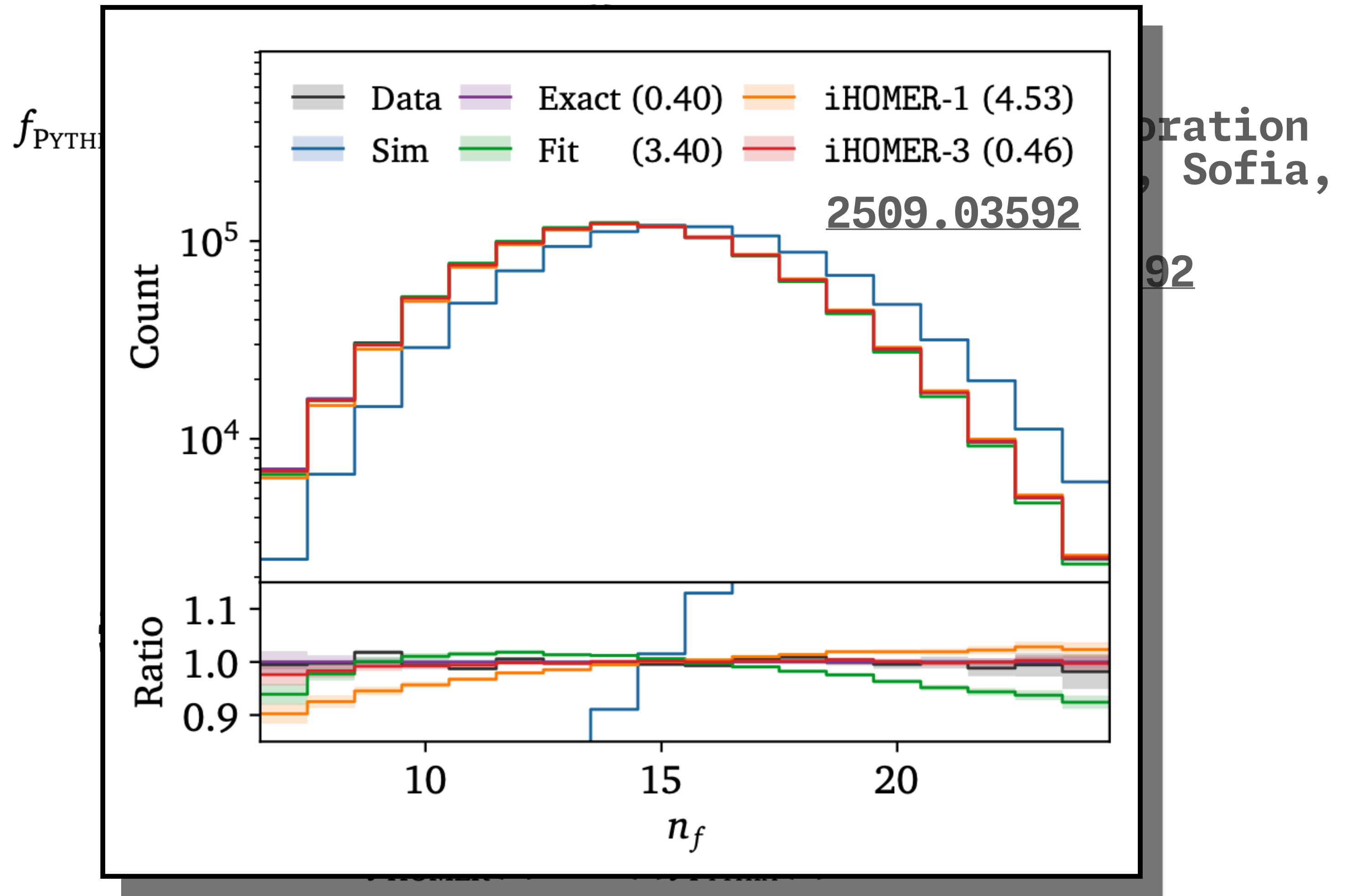
Training refinement
allowing for n
iterations of HOMER
“cycles” improves
fit convergence



Data-driven hadronization models

iHOMER

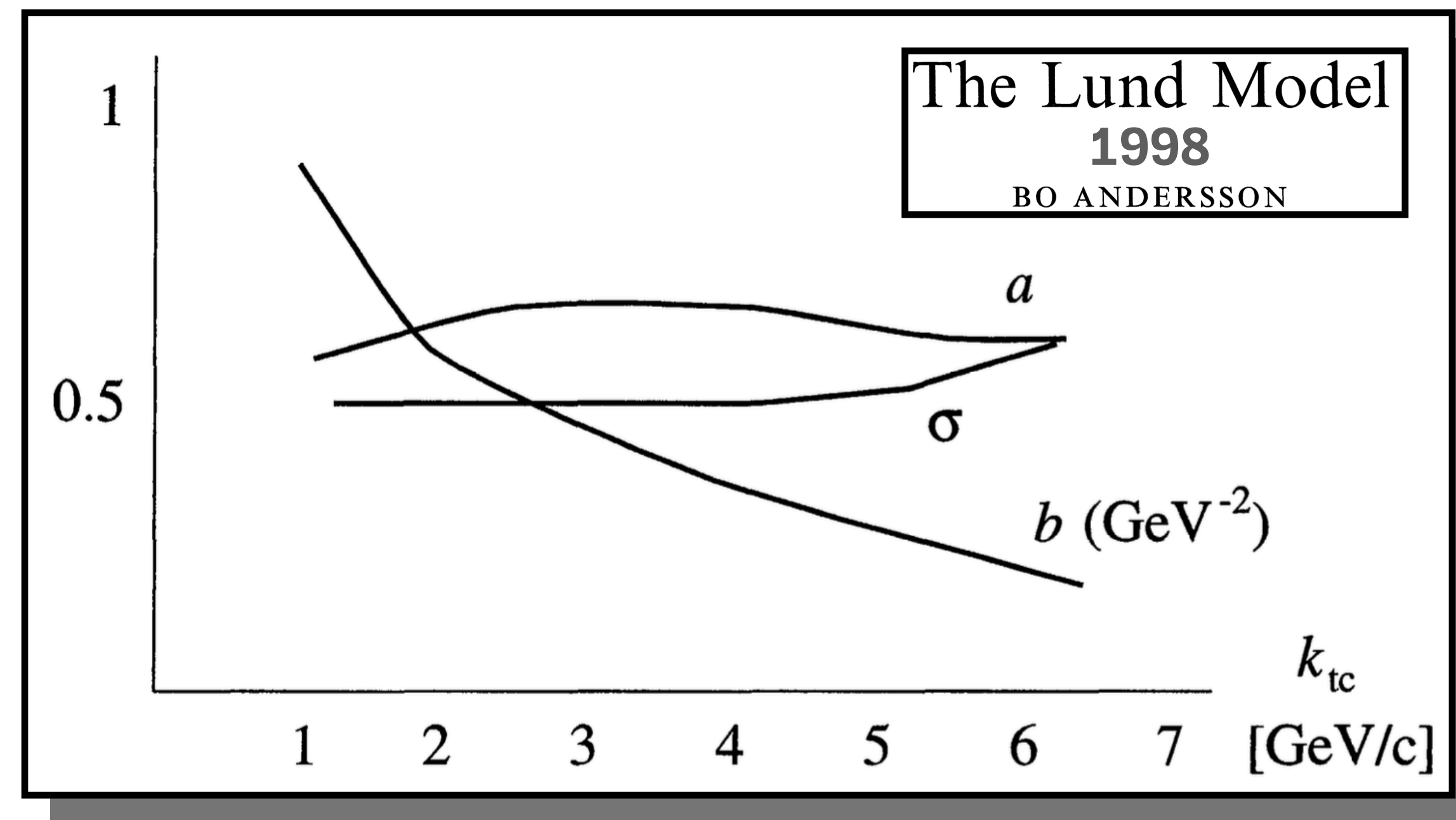
Training refinement
allowing for n
iterations of HOMER
“cycles” improves
fit convergence



Data-driven hadronization models

What about parton shower interplay?

How does the parton shower play into this game? Should I be worried that hadronization can absorb effects from the parton shower? **Some can!**



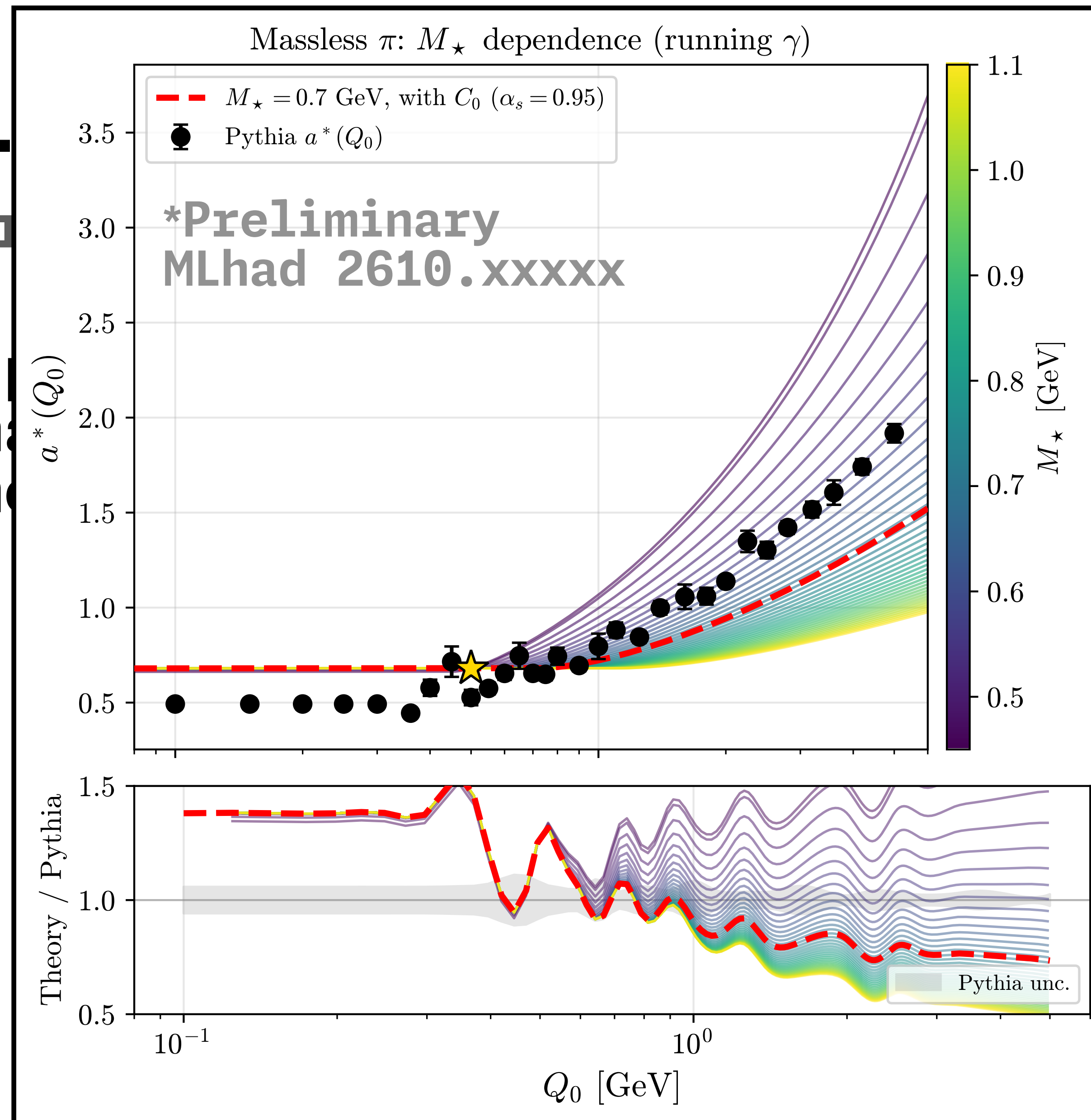
Data-driven

What about parton showers?

How does the parton shower
be worried that
the parton shower

models

time? Should I
effects from



Data-driven hadronization models

HOMER model compatibility analysis

What about different PS models?

HOMER is a powerful tool for investigating and understanding model compatibility!

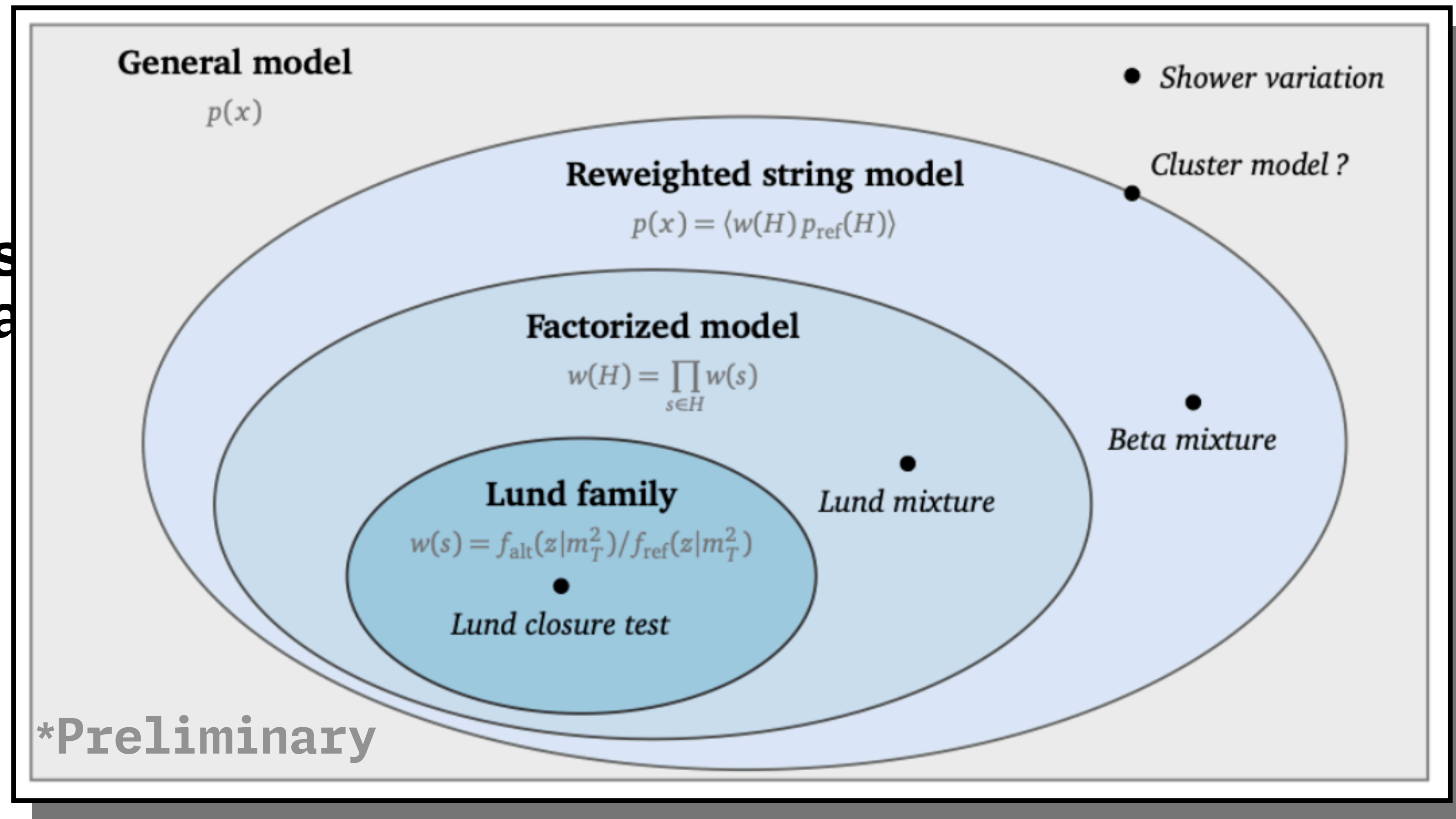
$$\mathcal{H}_{\text{string}}(\mathcal{P}_1) \stackrel{?}{=} \mathcal{H}_{\text{string}}(\mathcal{P}_2)$$

$$\mathcal{H}_{\text{string}}(\mathcal{P}_1) \stackrel{?}{=} \mathcal{H}_{\text{cluster}}(\mathcal{P}_1)$$

Data-driven hadronization models

HOMER model compatibility analysis

HOMER is
understand



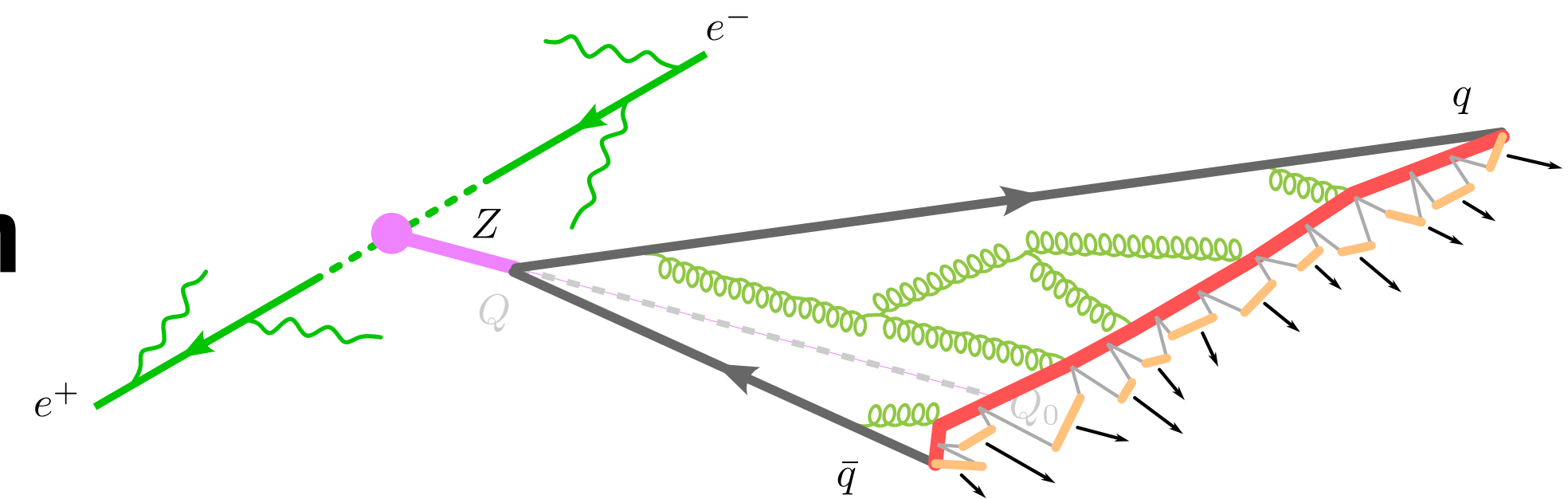
Upcoming work
in
collaboration
with Susie,
Ayo, Sofia,
Tilman + MLhad
26xx.xxxxx

Conclusions

Tuning data-driven hadronization models

MLHAD

- Data-driven hadronization models/methods are rapidly approaching “production” quality
- Flavor? Better observables? Train on data?
- Reweighting is pivotal to all of the methods described here! (Take it away Michael :))

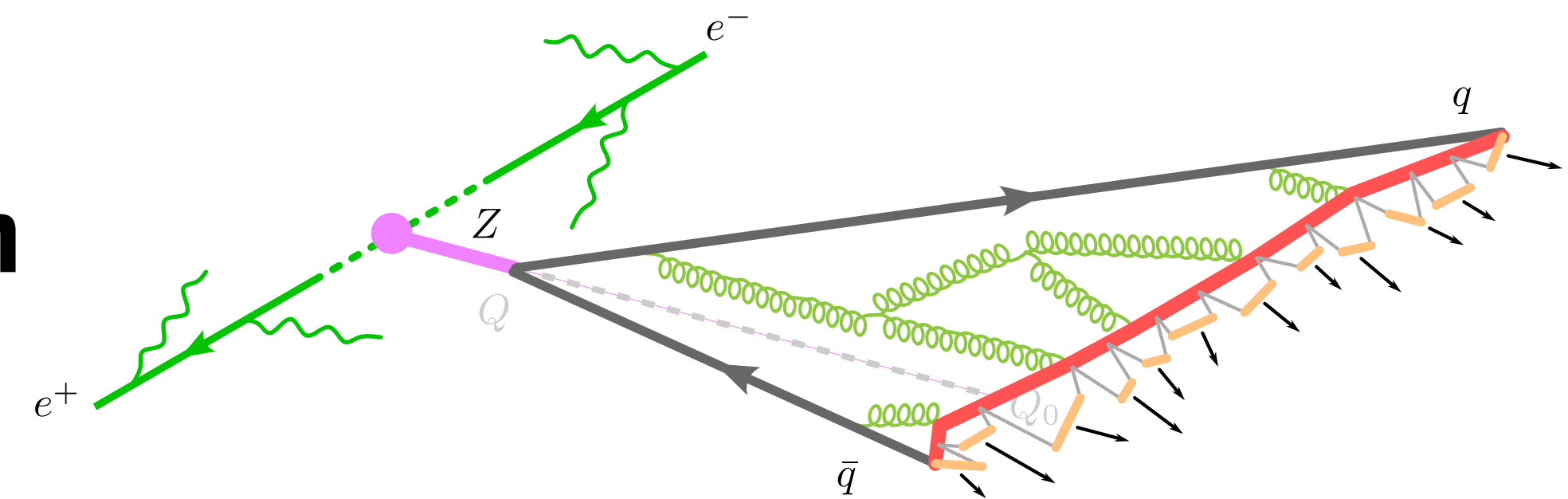


Conclusions

Tuning data-driven hadronization models

MLHAD

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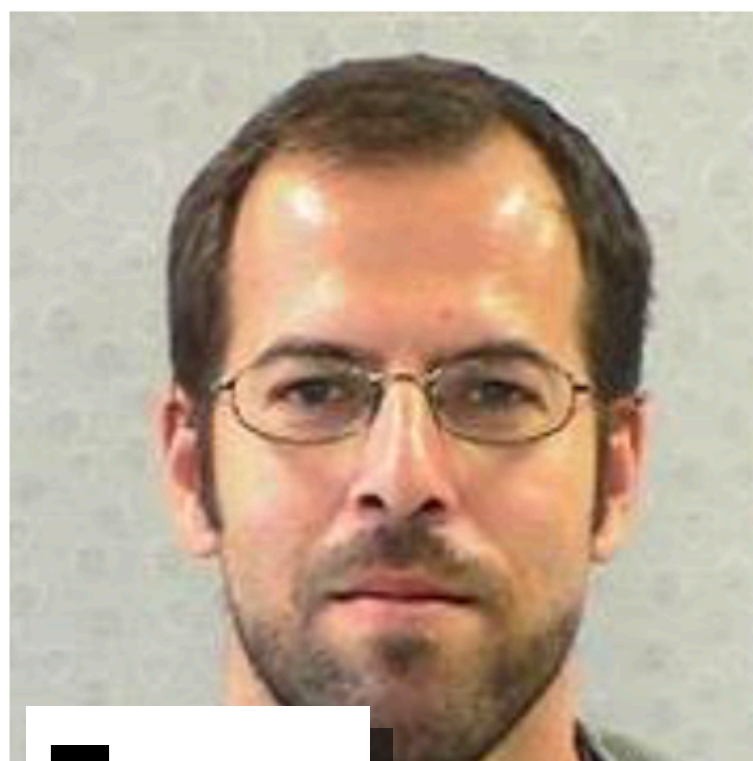


Thank you!

Back-ups

The squad

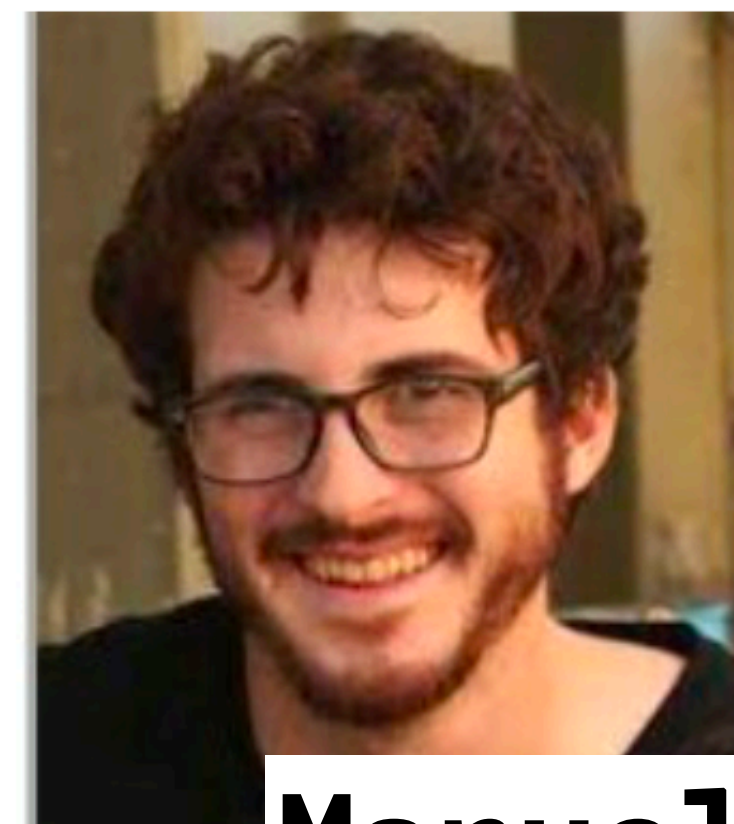
MILHEAD



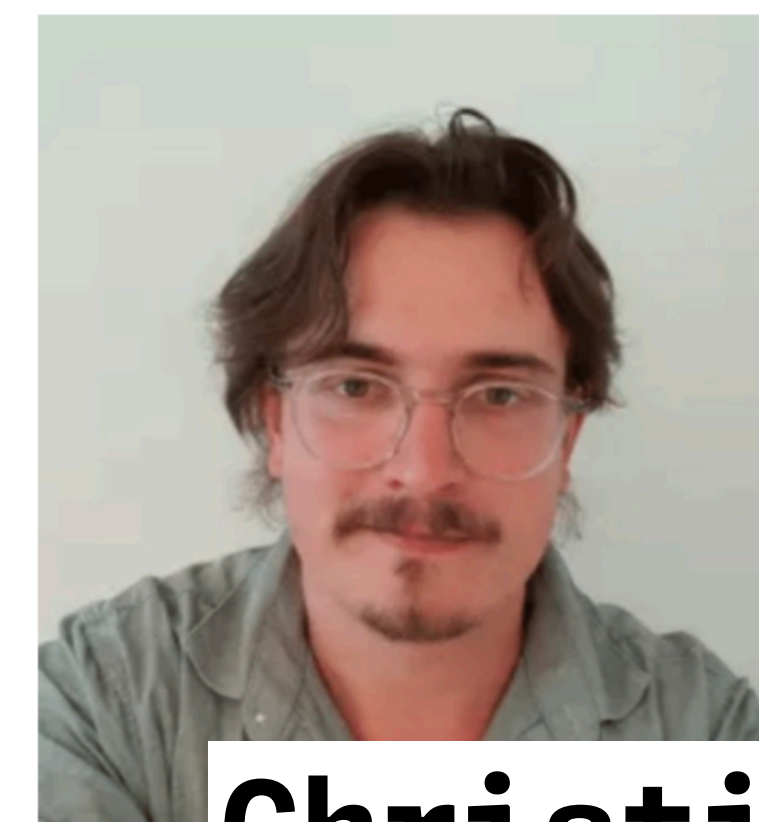
Jure



Phil



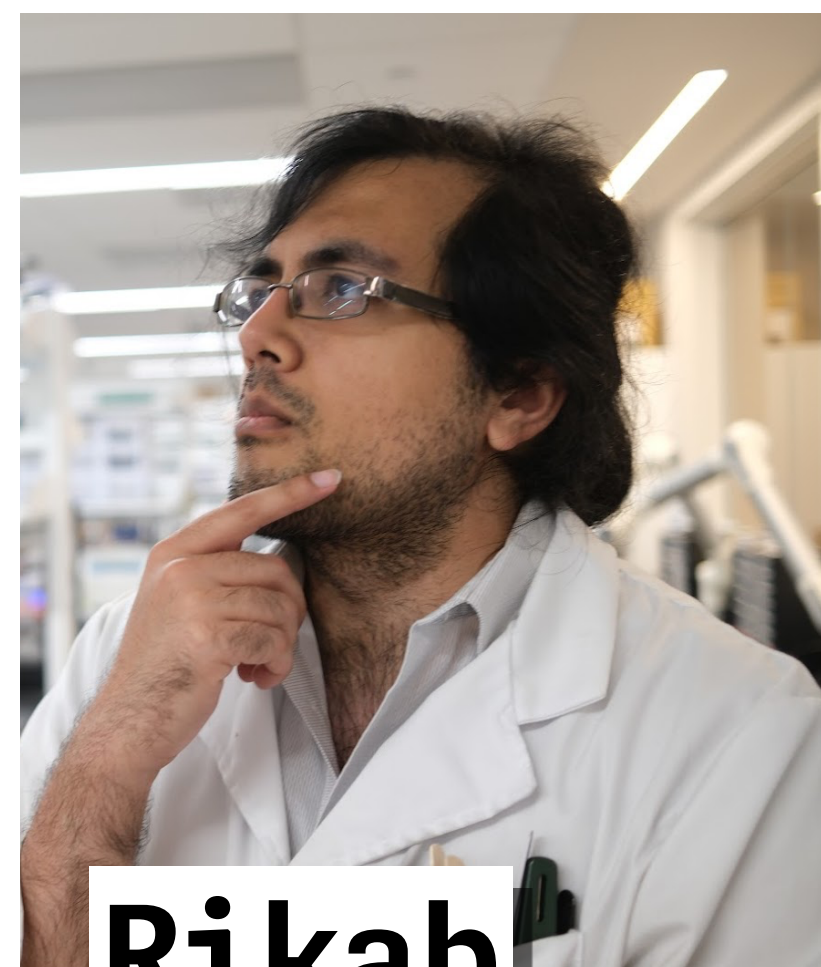
Manuel



Christian



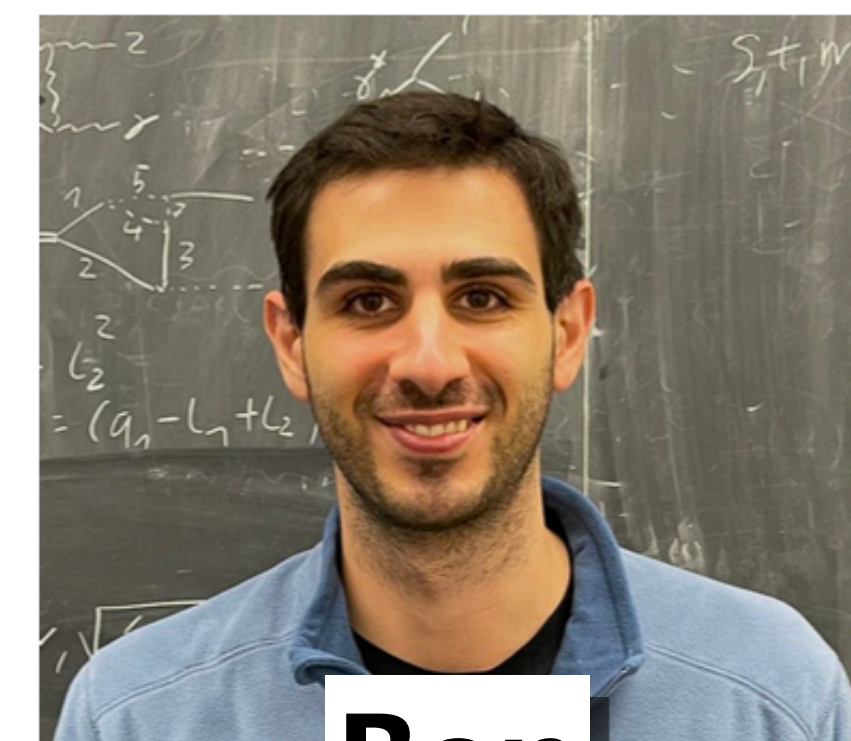
Steve



Rikab



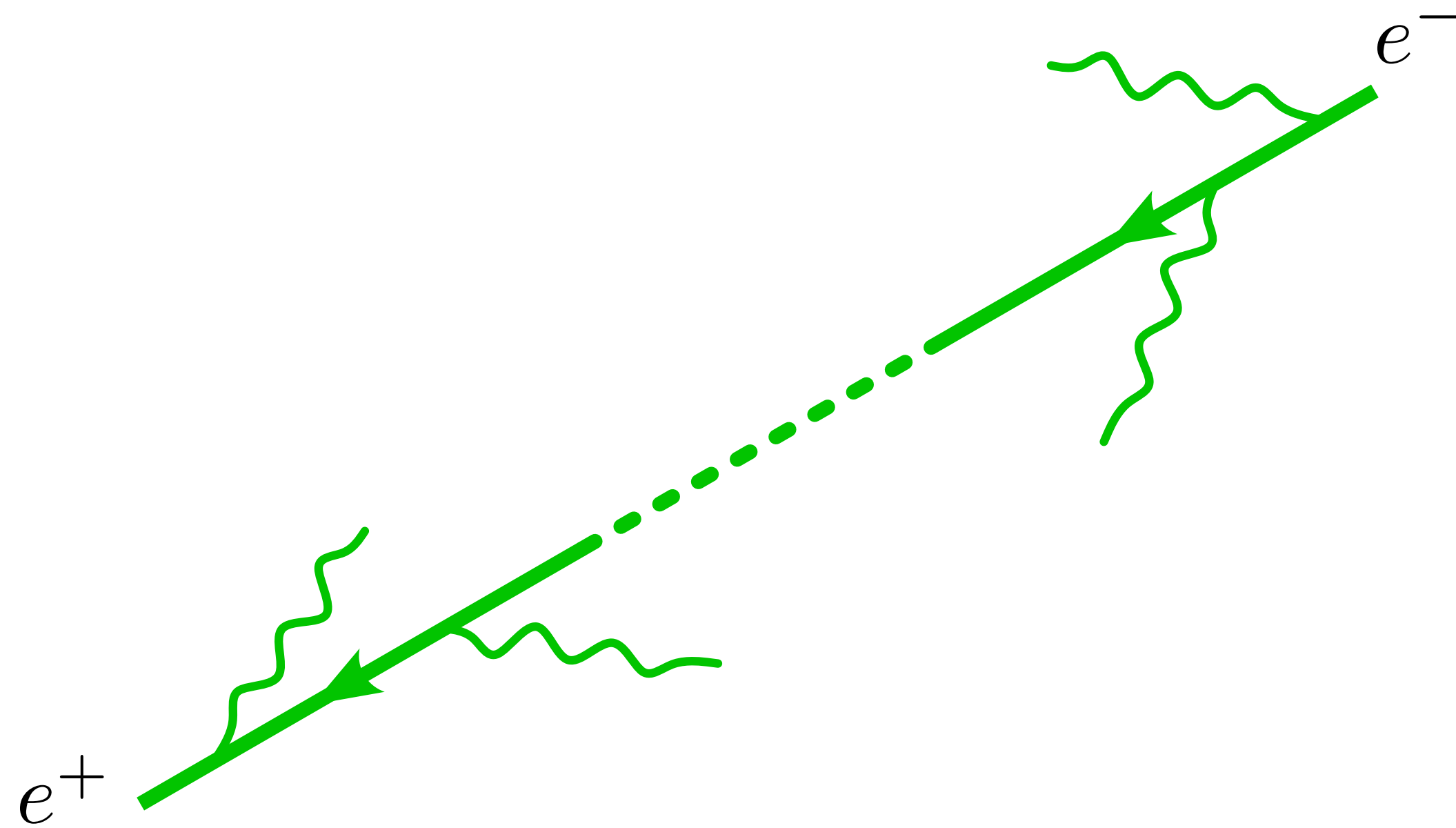
Michael



Ben

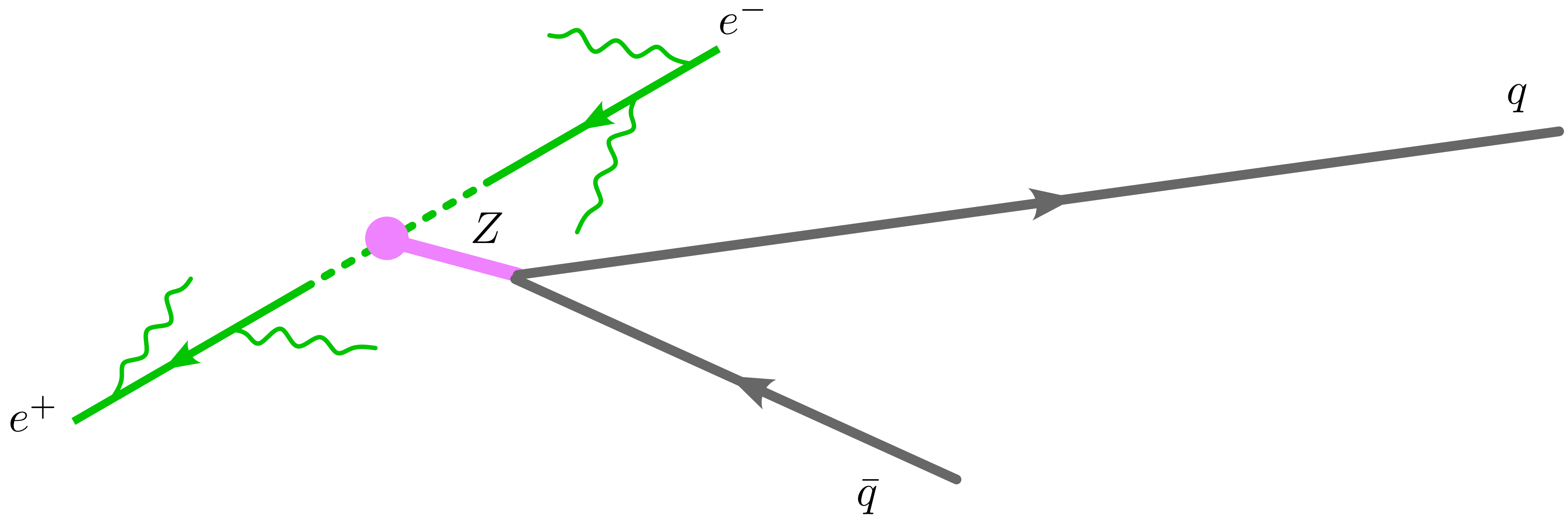
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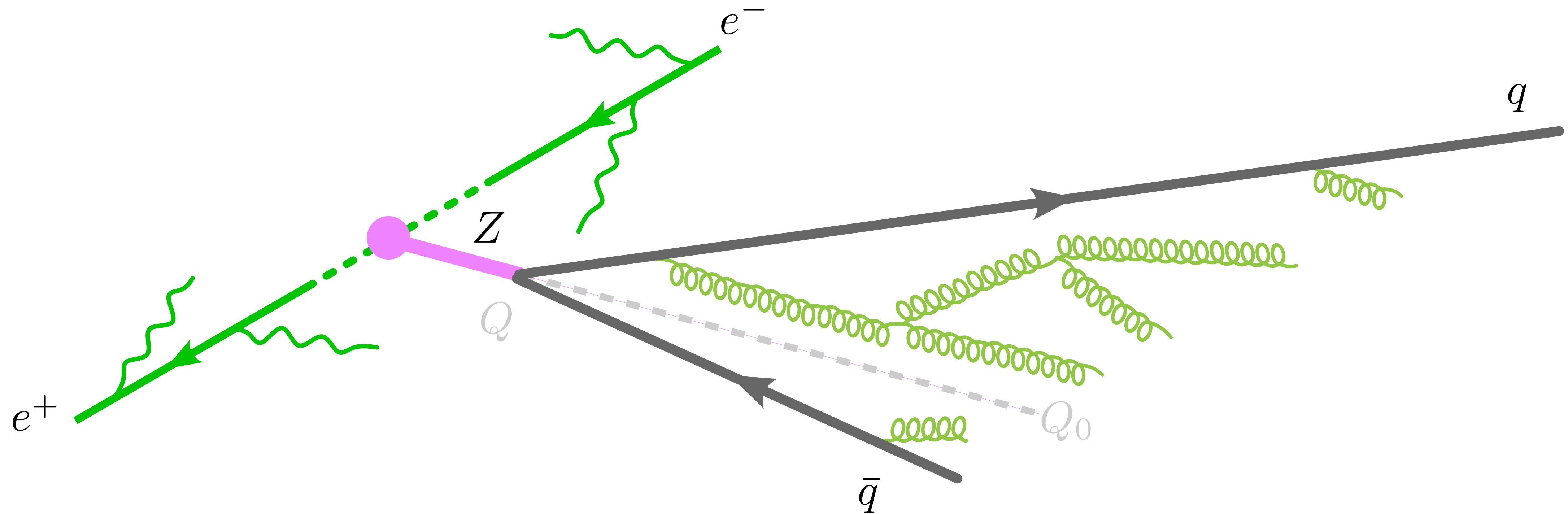
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